SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Accredited by NAAC A Grade

Assignment V

CO5: Analyze vulnerabilities in operating systems and applications and recommend hardening, defensive and reporting measures using data visualization and predictive modelling in R.

Evaluation note: This assignment rewards choosing, applying and defending the right visualization and model for a security question — not producing R output. Follow Import → Explore → Visualize → Model → Predict → Evaluate → Interpret → Recommend. Award marks for justified plot/model selection, correct metric-per-model-type, reliability/assumption checks and an evidence-based security decision. All data is authorized/lab only. Accept equivalent correct R.

Name of the Student : **SANA FATHIMA M**

Register No : **192371082**

Department : **CSE**

Sub Code / Name : **ITA1405 / ETHICAL HACKING**

Date of Submission : **12/11/2026**

Student Signature **Faculty Signature**

SANA FATHIMA M

ITA14 ETHICAL HACKING

ASSIGNMENT 5 – CO5

ETHICAL HACKING

Security-Data Visualization, Risk Modelling and Decision-Tree Evaluation for Vulnerability Analysis

Course Outcome 5:

CO5: Analyze vulnerabilities in operating systems and applications and recommend hardening, defensive and reporting measures using data visualization and predictive modelling in R.

CO5 Coverage

Visualization and Predictive Modelling in R – Graph creation (box plot, bar plot, histogram, line and scatter charts, heatmaps); regression (simple, multiple, logistic, Poisson); decision-tree modelling; and model evaluation applied to security-assessment data.

Nature: Individual

Duration: One Week

Total Marks: 50

Assessment Components

PART A – Analytical Modeling & Algorithm Development: 25 Marks

PART B – R-Based Implementation & Result Interpretation: 25 Marks

Assessment Design Philosophy

This assignment is designed for high-performing cybersecurity students. It emphasises the ability to choose, apply and defend the right visualization and model for a security question, not merely to produce R output.

PURPOSE OF THE ASSIGNMENT

Data visualization and predictive modelling provide powerful ways to understand the results of an authorized security assessment and to communicate risk. This assignment evaluates your ability to turn raw assessment data into visual evidence, risk models and defensible hardening recommendations, using R on authorized or lab data only.

This assignment investigates a fundamental question:

How should a security analyst use R to transform raw assessment data into meaningful insight and defensible hardening decisions?

You are expected to demonstrate the following progression:

Import → Explore → Visualize → Model → Predict → Evaluate → Interpret → Recommend

Simply generating a graph or obtaining model output is not sufficient. Your **R programming approach, choice of visualization, model selection, computational results, performance evaluation, and interpretation of findings** are all part of the assessment.

PART A – Analytical Modeling & Algorithm Development 25 Marks

Develop a clear and logical approach for solving the given security-data analysis and visualization problems.

TASK A1 – DESIGN A SECURITY-DATA ANALYSIS AND VISUALIZATION WORKFLOW |BTL6 – Create | K6

Integrated Visualization and Risk Modelling in R – 7 Marks

A security team maintains a dataset containing the results of authorized vulnerability assessments (hosts, OS type, open ports, patch age, hardening score, vulnerabilities found and severity).

You are not given the visualization or modelling technique to use. Based on the data and the security question, you must design the appropriate workflow.

A1.1 – Identify the Analytical Requirements | 1 Mark

Identify:

- Input data and relevant variables
- Type of variables involved
- Expected analytical outputs
- Visualization or modeling objective
- Key decision to be made

A1.2 – Select Appropriate Visualization Techniques | 1 Mark

For the given security dataset, determine suitable graphical representations for the key questions.

Consider appropriate use of:

- Box plot
- Bar plot
- Pie chart

- Histogram
- Line chart
- Scatter plot

Briefly justify why each selected visualization is appropriate for the security data.

A1.3 – Design the Visualization Workflow | 1 Mark

Develop a logical sequence for exploring and presenting the data using R.

Your workflow should indicate:

Data → Exploration → Visualization → Comparison → Observation

Explain how low-level plotting functions, ggplot2/lattice, heatmaps or other techniques fit into the workflow.

A1.4 – Design the Risk / Regression Analysis | 2 Marks

Determine an appropriate modelling approach based on the response variable (for example the severity score, the vulnerability count, or the compromised status).

Consider:

- Simple linear regression
- Multiple linear regression
- Logistic regression
- Poisson regression
- Decision-tree modeling

For the selected model, specify:

- Response variable
- Predictor variable(s)

- Reason for model selection
- Expected output
- Suitable performance/evaluation measure

Your answer should demonstrate why the selected model is appropriate for the security question.

A1.5 – Design the R Analysis Procedure | 1 Mark

Write structured, language-independent steps describing how you would:

Import → Explore → Visualize → Select Model → Fit Model → Evaluate → Interpret

The procedure should include the major analytical decisions and expected outputs without relying on a single predefined R function.

A1.6 – Interpret and Communicate the Results | 1

Mark Assume that the analysis produces graphs and

model results.

Explain how you would determine:

- Important patterns or relationships
- Significant differences or trends
- Model performance
- Practical conclusions from the analysis

Also indicate how the final graphs can be saved and presented appropriately in a security report.

Evidence Expected

Data understanding, visualization selection, analytical workflow, model design, R procedure and interpretation for the security assessment.

TASK A2 – OPTIMIZE A SECURITY-ANALYSIS WORKFLOW IN R

Advanced Visualization and Predictive Modelling in R – 5 Marks

A security operations team collects daily assessment data containing:

Scan_Date, OS_Type, Open_Ports, Patch_Age, Hardening_Score, Failed_Logins, Severity_Score,
Vulnerability_Count, Compromised, Service_Count

An analyst proposes the following R workflow:

Import the dataset

Create a histogram for every variable

Create a 3D graph for all variables

Apply simple linear regression

Use the same model to predict every target variable

Display the results

The team wants to use the analysis to identify vulnerability patterns, the main risk drivers and the hosts most likely to be compromised.

A2.1 – Identify Limitations in the R Analysis| 1 Mark

Identify **two major analytical problems** in the proposed workflow.

Explain why blindly applying the same visualization and modelling technique to every variable is inappropriate for security data.

A2.2 – Redesign the Visualization Strategy | 1

Mark Develop an improved R visualization

strategy for the dataset.

Select suitable techniques for:

- Distribution of Severity_Score
- Vulnerability comparison across OS_Type
- Relationship between Patch_Age and Vulnerability_Count
- Vulnerability trend over Scan_Date
- Correlation among the numerical security metrics

Your solution should consider the appropriate use of box plot, bar plot, histogram, line chart, scatter plot and heatmap.

A2.3 – Design an Appropriate Predictive Modeling Strategy | 2 Marks

The company has three objectives:

1. Predict Severity_Score using Open_Ports, Patch_Age, Hardening_Score and Failed_Logins.
2. Predict whether a host will be Compromised = Yes/No.
3. Predict the number of vulnerabilities found on a host.

Select an appropriate model for each objective from:

Simple Linear Regression → Multiple Linear Regression → Logistic Regression → Poisson Regression → Decision Tree

For each selection, provide a technical justification based on the response variable and the data.

A2.4 – Evaluate and Challenge the Model | 1 Mark

After fitting the selected models, the analyst obtains poor prediction performance for one model.

Propose **two R-based actions** that could be used to investigate or improve the model.

Your answer may consider residual analysis, variable selection, model type, data quality or class imbalance.

Evidence Expected

Diagnosis of the weak R workflow, a redesigned visualization strategy, a justified predictivemodelling strategy and improvement actions for the security data.

TASK A3 – EVALUATE VISUALIZATION TECHNIQUES IN

R|BTL5 – Evaluate | K5 5 Marks

Consider the following analytical requirements for a Vulnerability Assessment Dataset:

1. Distribution of Severity Scores
2. Comparison of Vulnerabilities across OS Types
3. Relationship between Patch Age and Vulnerabilities
4. Monthly Vulnerability Trend
5. Correlation among the Numerical Security Metrics

Complete the following analysis:

Analytical Requirement	**Recommended Visualization**	**R Variable Type**	**Why is it Suitable?**	**Alternative Technique**
Distribution of Severity Scores				
OS-wise Vulnerabilities				
Patch Age vs				

Vulnerabilities Monthly Vulnerability			
Trend			
Metric Correlation			

A3.1 – Evaluate the Visualization Choices | 2 Marks

Evaluate the suitability of the different R visualization techniques for these security requirements.

Consider:

Box Plot → Bar Plot → Histogram → Line Chart → Scatter Plot → Heatmap

Justify why a particular visualization is more appropriate than an alternative for each requirement.

A3.2 – Compare R Visualization Approaches | 2 Marks

Compare **Base R plotting, ggplot2, and lattice** with respect to:

- Ease of customization
- Handling of multiple variables
- Presentation quality
- Suitability for complex datasets

Recommend the most appropriate approach for the given vulnerability dataset and justify it.

A3.3 – Evaluate the Need for Advanced Graphics | 1 Mark

A student proposes using a **3D graph and interactive graph** for every analysis.

Evaluate this proposal and identify one situation where advanced or interactive graphics genuinely help in security analysis.

Evidence Expected

A completed comparison, justified visualization choices, an evaluation of the R approaches and a recommendation for the security data.

TASK A4 – EVALUATE RISK AND PREDICTIVE MODELS IN R|BTL5 – Evaluate | K5

Simple Linear, Multiple, Logistic, Poisson Regression and Decision Trees in

Security Analysis A security team wants to analyze host and assessment data using

R. The dataset contains:

Vulnerability_Count, Patch_Age, Open_Ports, Hardening_Score, Compromised, Failed_Logins, Severity_Score

Consider the following analytical objectives:

1. Predict Severity_Score using Patch_Age.
2. Predict Severity_Score using Patch_Age, Open_Ports and Hardening_Score.
3. Predict whether a host is Compromised (Yes/No).
4. Predict the Number of Vulnerabilities on a host.
5. Predict whether a host is Compromised (Yes/No) using a decision tree.

For each objective, prepare a concise analysis:

A4.1 – Evaluate Model Suitability | 2 Marks

Simple Linear Regression → Multiple Linear Regression → Logistic Regression → Poisson Regression → Decision Tree

for the given objectives.

Justify the selected model based on the **nature of the response variable and the purpose of prediction**.

A4.2 – Compare Predictive Models | 1 Mark

Suppose both Logistic Regression and a Decision Tree are used to predict whether a host is compromised.

Identify two criteria that can be used to compare their performance and explain them.

A4.3 – Evaluate Model Results | 1 Mark

Suppose a model produces good training performance but poor test performance.

Evaluate the possible reason for this behaviour and recommend one suitable

remedy. **Inquiry**

Answer:

Why is selecting a model based only on its prediction accuracy insufficient for a security decision?

Evidence Expected

Correct model selection, comparison of alternative models, evaluation of results and reasoning for the security context.

TASK A5 – ANALYZE WHEN A SIMPLE R MODEL MAY BE INSUFFICIENT BTL4 – Analyze | K4

Model Selection and Complexity in Security Analysis in R – 4 Marks

A security team wants to predict the Severity_Score of a host using the following variables:

Patch_Age, Open_Ports, Hardening_Score, Failed_Logins, Service_Count

The analyst proposes using **Simple Linear Regression** for all prediction tasks because it is easy to implement and interpret.

However, the team wants to determine whether a more suitable model should be used.

A5.1 – Simple Model Perspective | 1 Mark

Explain how Simple Linear Regression can be used to predict Severity_Score using one predictor.

Evaluate why using only one predictor may become inadequate when several factors affect the severity.

A5.2 – Alternative Model Design | 2 Marks

Design a structured R-based analytical approach for selecting between:

Simple Linear Regression → Multiple Linear Regression → Decision Tree

Your approach should explicitly consider:

- Response variable ● Predictor variables
- Model assumptions
- Model performance
- Model comparison
- Final model selection

The procedure should indicate how the models can be **fitted, evaluated, compared, and selected** in R.

A5.3 – What Have We Sacrificed? | 1 Mark

Explain the distinction between:

Obtaining a model with good predictive

performance and

Selecting a model that is appropriate, interpretable and reliable for security decisions

Discuss why choosing a model based only on prediction accuracy may lead to poor security decisions.

Evidence Expected

Model-selection reasoning, a structured R analytical procedure, a comparison and justification for the security context.

PART B – IMPLEMENTATION & COMPUTATIONAL INVESTIGATION

25 Marks

Part B assesses whether your R-based analytical decisions produce meaningful, correct and defensible results on security-assessment data.

You are expected to:

Implement → Visualize → Model → Evaluate → Compare → Interpret

Simply displaying graphs or R output is insufficient. Your code, analysis and interpretation must support a security decision.

TASK B1 – IMPLEMENT AND EVALUATE AN R VISUALIZATION WORKFLOW| BTL6 – Create | K5

Graph Creation and Advanced Visualization – 7 Marks

Use a suitable vulnerability-assessment dataset containing numerical, categorical and date variables.

Implement an R-based visualization workflow covering the appropriate graphical techniques from:

Box Plot → Bar Plot → Pie Chart → Histogram → Line Chart → Scatter Plot

Your program should also demonstrate appropriate use of **ggplot2 or lattice**, and where relevant, **heatmaps, interactive graphics, or 3D visualization**.

B1.1 – Prepare and Explore the Data | 2 Marks

Import and inspect the dataset using R.

Perform suitable preprocessing and identify:

- Numerical and categorical variables
- Missing or inconsistent values
- Variables suitable for graphical analysis
- Relationships that require further investigation

Explain the preprocessing decisions made.

B1.2 – Implement the Visualization Set | 2 Marks

Create appropriate graphs to answer at least **five different analytical questions**.

For example:

- What is the distribution of severity scores?
- How do vulnerabilities vary across OS types?
- Is there a trend in vulnerabilities over time?
- Is there a relationship between patch age and vulnerabilities?
- Which numerical security metrics show strong correlation?

Each graph must include appropriate **title, axis labels, scale, legend, and other relevant presentation elements**.

B1.3 – Compare Visualization Approaches | 1 Mark

Implement at least one visualization using **Base R** and the equivalent visualization using **ggplot2 or lattice**.

Compare the approaches based on:

- Readability
- Customization
- Ease of interpretation
- Suitability for the dataset

B1.4 – Evaluate the Visualization Results | 2 Marks

Analyze the generated graphs and identify:

- Important patterns
- Outliers
- Trends
- Relationships
- Unexpected observations

Explain how the visual evidence could influence a subsequent modelling or hardening decision.

Evidence Expected

R source code, dataset preparation, multiple visualizations, a comparison and interpreted findings for the security data.

TASK B2 – ENGINEER AN INFORMATIVE VISUALIZATION| BTL5 – Evaluate | K5

Same Data, Different Visual Conclusions – 5 Marks

Use the dataset and visualizations developed in **Task B1**.

Construct two different visualization strategies for the **same analytical question**.

Strategy A

Use a conventional Base R visualization.

Strategy B

Use an appropriately designed **ggplot2/lattice/interactive visualization**.

Before executing, predict:

- Which visualization will communicate the pattern more clearly

- Which variables or categories may dominate the display
- Where misleading interpretation may occur

Then implement both strategies.

B2.1 – Compare the Visualizations | 2 Marks

Compare the two approaches based on:

Measure	Strategy A	Strategy B
Readability		
Information conveyed		
Outlier visibility		
Ease of comparison		
Customization		

B2.2 – Investigate Visualization Quality | 1 Mark

Modify at least **two graphical parameters** such as:

- Scale
- Labels
- Breaks
- Grouping
- Faceting
- Transformation
- Legend

Explain how the modification changes the interpretation.

B2.3 – Challenge the Visualization | 1 Mark

Identify one situation in which a 3D or interactive graph could provide genuine value in security analysis.

B2.4 – Analytical Decision | 1 Mark

Based on your evidence, recommend the better visualization approach for the security question.

Justify your recommendation.

Evidence Expected

Two visualization strategies, a prediction, the R implementation, a comparison and a justified recommendation.

TASK B3 – IMPLEMENT AND COMPARE RISK MODELS | BTL5 – Evaluate | K5

Regression and Predictive Modelling in R – 8 Marks

Use a suitable security dataset containing variables that support different types of models.

Develop R implementations for:

- **Simple Linear Regression**
- **Multiple Linear Regression**
- **Logistic Regression**
- **Poisson Regression**

You may additionally use a **Decision Tree** for comparison.

B3.1 – Build and Validate the Models | 2 Marks

Implement the appropriate models for the selected response variables (for example the severity score, the vulnerability count and the compromised status).

For each model, report:

- Response variable

- Predictor variables
- Estimated model parameters
- Important statistical results
- Predictions

Explain why the selected model is appropriate for the response variable.

B3.2 – Evaluate Model Performance | 2

Marks Use suitable performance

measures for each model.

Examples include:

- **R^2 / Adjusted R^2**
- **RMSE / MAE**
- **Confusion Matrix**
- **Accuracy**
- **Precision / Recall**
- **Deviance**
- **AIC**

Do not use a single performance measure for all model types.

B3.3 – Compare Alternative Models | 2 Marks

For at least one prediction problem, compare two competing approaches, such as:

Logistic Regression vs Decision Tree

or

Simple Linear Regression vs Multiple Linear

Regression Prepare an evidence table:

Model Evaluation Measure Training Result Test Result Interpretation

Model 1

Model 2

Determine which model performs better and justify your decision.

B3.4 – Investigate Model Reliability | 2

Marks Investigate whether the selected model

is reliable.

Consider:

- Residual analysis
- Outliers
- Overfitting
- Predictor significance
- Multicollinearity
- Train/test performance
- Model assumptions

Identify at least one limitation of your selected model and recommend an improvement.

Evidence Expected

R source code, model outputs, predictions, performance measures, a comparison and a reliability analysis for the security models.

TASK B4 – FINAL SECURITY ANALYTICS DECISION AND ANALYSIS| BTL4 – Analysis | K5

Visualization and Predictive Modelling for a Real Security Decision – 5 Marks

A security team wants to use R to determine whether a new hardening programme reduces risk, and to predict which hosts are likely to be compromised.

The dataset contains:

Scan_Date, Severity_Score, Patch_Age, Hardening_Score, Open_Ports, OS_Type, Failed_Logins, Vulnerability_Count, Compromised

The management requires:

1. Understanding vulnerability patterns
2. Identifying important relationships
3. Predicting future risk and severity
4. Predicting host-compromise behaviour
5. Selecting an appropriate analytical approach for future deployment **B4.1 –**

Implement an End-to-End R Strategy | 2 Marks

Develop an R workflow covering:

Data Preparation → Visualization → Model Selection → Model Training → Evaluation → Recommendation

Your output must include appropriate graphs and predictive results.

B4.2 – Challenge Your Analytical Strategy | 1 Mark

Construct or identify a situation in which your selected model produces misleading results.

Investigate the possible reason and demonstrate one modification in R that addresses it.

B4.3 – Final Analytical Decision | 2 Marks

Prepare a short Security Analytics Decision Note addressing:

- Which visualization provides the most useful security insight?
- Which predictive model performs best?

- What evidence supports your selection?
- What are the limitations of the selected approach?
- Would you recommend deploying the model for security decision-making?

Your conclusion must be based on **experimental evidence rather than only theoretical reasoning**.

Evidence Expected

A complete R implementation, visualization outputs, predictive results, an evaluation and a final security decision note.

.

MARK DISTRIBUTION

Question	Assessed Area	Marks	BTL	OBE Knowledge
A1	Design a Security-Data Analysis and Visualization Workflow	7	BTL6	K5
A2	Optimize a Security-Analysis Workflow in R	5	BTL5	K5
A3	Evaluate Visualization Techniques in R	5	BTL5	K5
A4	Evaluate Risk and Predictive Models in R	4	BTL5	K5
A5	Analyze When a Simple Model May Be Insufficient	4	BTL4	K4
B1	Create and Evaluate an R Visualization Workflow	7	BTL6	K5
B2	Engineer an Informative Visualization	5	BTL5	K5
B3	Implement and Compare Risk Models	8	BTL5	K5
B4	Final Security Analytics Decision	5	BTL4	K5

TOTAL		50		

Submission Requirements

For Part A (A1–A5), submit the required problem formulation, analytical design and justification.

For Part B (B1–B4), submit the R source code together with the dataset used.

Students are expected to follow the same analytical engineering workflow throughout the assignment:

UNDERSTAND → DESIGN → VISUALIZE → MODEL → IMPLEMENT → MEASURE → CHALLENGE → DECIDE

The key question is not simply:

"Did the R program produce an output?"

It is also: "Why was this visualization or model selected, how effective is it, and what security decision does it support?"

Evidence Requirements

Students should provide evidence appropriate to each task, including:

- Problem formulation and analytical objectives
- Data understanding and variable identification
- Visualization and model selection with justification
- Structured pseudocode or analytical procedure
- R implementation and reproducible source code
- Graphs and visualization outputs
- Regression/model summaries and predictions
- Appropriate evaluation metrics
- Comparison of alternative visualization/modeling approaches

- Identification of assumptions, limitations, outliers, or poor performance
- Interpretation of results and final analytical decisions

The submission should demonstrate **reasoning, implementation, measurement, evaluation, and interpretation**, rather than reporting final outputs alone.

TABLE OF CONTENTS

1. Introduction, Objectives and Dataset
2. Reproducibility and Execution Note
PART A – Analytical Modeling and Algorithm Development (25 Marks)
Task A1 – Design a Security-Data Analysis and Visualization Workflow (7)
Task A2 – Optimize a Security-Analysis Workflow in R (5)
Task A3 – Evaluate Visualization Techniques in R (5)
Task A4 – Evaluate Risk and Predictive Models in R (4)
Task A5 – When a Simple R Model May Be Insufficient (4)
PART B – Implementation and Computational Investigation (25 Marks)
Task B1 – Implement and Evaluate an R Visualization Workflow (7)
Task B2 – Engineer an Informative Visualization (5)
Task B3 – Implement and Compare Risk Models (8)
Task B4 – Final Security Analytics Decision and Analysis (5)
7. Expected Deliverables and Outcomes Summary
8. Conclusion
Appendix A – Dataset Dictionary and Complete R Script

1. INTRODUCTION, OBJECTIVES AND DATASET

Data visualization and predictive modelling convert raw results of an authorised vulnerability assessment into visual evidence, quantified risk and defensible hardening recommendations. This report follows the workflow required by the assignment: **Import → Explore → Visualize → Model → Predict → Evaluate → Interpret → Recommend**. The emphasis is on why each visualization and model is chosen, how well it performs, and which security decision it supports.

1.1 Objectives

- Understand vulnerability patterns across operating systems, patch age and time.
- Identify the main risk drivers of severity, vulnerability count and host compromise.
- Select, fit, evaluate and compare regression and decision-tree models using metrics that suit each response type.
- Judge whether a hardening programme reduces risk and whether a compromise-prediction model is fit for deployment.

1.2 Dataset used

The file **vuln_assessment.csv** (submitted with this report) contains **400 hosts** from an authorised lab assessment between January and December 2025. A hardening programme started on 1 July 2025 (variable Hardening_Programme = 1). Data was randomly split 70% training (280 hosts) and 30% test (120 hosts).

Variable	Type	Role in analysis
host_id	Identifier (character)	Not analysed
Scan_Date	Date	Trend analysis (line chart)
OS_Type	Categorical (4 levels)	Group comparison, facet
Open_Ports, Failed_Logins, Service_Count	Numeric (discrete)	Predictors
Patch_Age (days)	Numeric (continuous)	Predictor
Hardening_Score (0–100)	Numeric (continuous)	Predictor
Severity_Score (0–10)	Numeric (continuous)	Response of linear regression
Vulnerability_Count	Count (non-negative	Response of Poisson

	integer)	regression
Compromised (Yes/No)	Binary categorical	Response of logistic regression / tree
Hardening_Programme (0/1)	Binary categorical	Pre/post comparison (t-test)

1.3 Security question

Which hosts and operating systems should be hardened first, what drives severity, vulnerability count and compromise, does the hardening programme reduce risk, and can a model reliably flag hosts likely to be compromised?

2. REPRODUCIBILITY AND EXECUTION NOTE

The R source code in every Part B task is complete and runs on the submitted CSV in RStudio. The console and graph screenshots in this report were produced from the same dataset using equivalent computations (ordinary least squares, binomial and Poisson GLMs, a depth-3 classification tree and the same plot types), because R itself was not installed in the environment used to prepare this document. Because R's sample() uses a different random generator, a train/test split run in RStudio will give slightly different numbers. For final submission, re-run the code in RStudio, and replace the screenshots if your marker requires native RStudio captures. Code listings use Courier New for readability; all other text is Times New Roman 12 pt.

PART A – ANALYTICAL MODELING AND ALGORITHM DEVELOPMENT (25 MARKS)

TASK A1 – Design a Security-Data Analysis and Visualization Workflow (7 Marks)

Scenario: a security team keeps results of authorised vulnerability assessments (hosts, OS type, open ports, patch age, hardening score, vulnerabilities found and severity). The technique is not given; the workflow must be designed.

Question A1.1 (1 Mark): Identify the input data and relevant variables, variable types, expected analytical outputs, visualization/modelling objective and the key decision to be made.

Answer:

Aspect	Answer
Input data	Authorised vulnerability-assessment records, one row per host per scan: host_id, Scan_Date, OS_Type, Open_Ports, Patch_Age, Hardening_Score, Failed_Logins, Service_Count, Vulnerability_Count, Severity_Score, Compromised.
Variable types	Categorical: OS_Type, Compromised. Numeric continuous: Patch_Age, Hardening_Score, Severity_Score. Count (discrete): Open_Ports, Failed_Logins, Service_Count, Vulnerability_Count. Date: Scan_Date. Identifier: host_id.
Expected outputs	Distribution of severity; group comparison across OS; relationship and trend plots; correlation matrix; a severity regression with coefficients; a count model; a compromise probability for each host.
Objective	Find the drivers of vulnerability and severity and flag high-risk hosts (visual exploration followed by predictive modelling).
Key decision	Which hosts and which OS families should be patched and hardened first, and whether a model may be used as a decision-support flag.

Question A1.2 (1 Mark): Select suitable graphical representations for the key questions and justify each.

Answer:

Question	Chosen plot	Justification
Distribution of severity	Histogram (with box plot)	One continuous variable; histogram shows shape, skew and modes; box plot shows median, quartiles and outliers.
Vulnerabilities across OS	Grouped box plot (bar for host counts)	Compares a numeric variable across categories while showing spread and outliers; a bar chart is used only for counts.
Patch age vs vulnerabilities	Scatter plot with trend line	Both variables are numeric; shows direction, strength and outliers of the relationship.
Vulnerabilities over time	Line chart	Scan_Date is ordered, so a line shows trend and seasonality, including the effect of the hardening start date.
Share compromised	Pie chart (single share only)	A pie chart is acceptable only for one share-of-total with two or three slices; angles are hard to compare for many categories.
Correlation among metrics	Heatmap	Displays many numeric–numeric correlations at once and reveals multicollinearity.

Question A1.3 (1 Mark): Develop a logical sequence for exploring and presenting the data in R (Data → Exploration → Visualization → Comparison → Observation) and explain where low-level plotting, ggplot2/lattice and heatmaps fit.

Answer:

- **Data:** read.csv(), set factors (OS_Type, Compromised) and Date (Scan_Date).
- **Exploration:** str(), summary(), colSums(is.na()), and boxplot.stats() to find missing, inconsistent and outlying values.

- **Visualization:** univariate first (hist, boxplot) then bivariate (scatter, grouped box, line). Low-level Base R functions (hist, plot, abline, boxplot, lines, legend) are used for quick checks while exploring.
- **Comparison:** ggplot2 with facet_wrap() and colour aesthetics compares OS families, compromised vs not compromised, and pre- vs post-hardening periods; lattice (xyplot, bwplot) is a valid alternative with conditioning panels.
- **Heatmap:** corrplot()/ggplot2 geom_tile() for the correlation matrix, which guides predictor selection.
- **Observation:** write down patterns, outliers and trends and convert them into hypotheses for modelling and hardening.

Question A1.4 (2 Marks): Determine an appropriate modelling approach based on the response variable; specify response, predictors, reason, expected output and evaluation measure.

Answer:

Item	Selected design
Selected model	Multiple linear regression (primary, for severity). Logistic regression and Poisson regression are used for the binary and count responses.
Response variable	Severity_Score (continuous, 0–10).
Predictors	Patch_Age, Open_Ports, Hardening_Score, Failed_Logins.
Reason	The response is continuous and several plausible risk drivers act together; multiple regression estimates the independent effect of each driver while adjusting for the others, and the coefficients are directly interpretable as hardening actions.
Expected output	Coefficient estimates with p-values (e.g. severity change per extra day of patch age), fitted/predicted severity per host.
Evaluation	Adjusted R^2, RMSE/MAE on a test set, residual diagnostics, VIF for multicollinearity.
Other responses	Compromised (Yes/No) → logistic regression (accuracy, precision/recall, AUC); Vulnerability_Count → Poisson regression (deviance, AIC, dispersion); decision tree as an interpretable comparator.

Question A1.5 (1 Mark): Write structured, language-independent steps: Import → Explore → Visualize → Select Model → Fit Model → Evaluate → Interpret.

Answer:

1. **Import:** load the file; declare each column as numeric, category or date. Output: a typed table.
2. **Explore:** list types, ranges, missing and impossible values; decide treatment (impute with median, correct or remove). Output: a clean table and a data-quality log.
3. **Visualize:** draw univariate then bivariate graphs; compute the correlation matrix. Output: patterns, outliers, candidate predictors.
4. **Select model:** decide by response type: continuous → linear; binary → logistic; count → Poisson; rule-based non-linear → tree. Output: one justified model per question.
5. **Fit model:** split data into training and test sets with a fixed random seed; estimate parameters on training data only. Output: coefficients and standard errors.
6. **Evaluate:** measure fit and prediction on the test set using the metric suited to the model; check assumptions and overfitting. Output: performance table and diagnostics.
7. **Interpret:** link each significant driver to a hardening action and state limitations. Output: recommendations and a decision note.

Question A1.6 (1 Mark): Explain how important patterns, significant differences, model performance and practical conclusions are determined, and how graphs are saved and presented in a security report.

Answer:

- **Patterns/relationships:** the shape of histograms (skew), the slope and spread of scatter plots, and the strength of correlation coefficients in the heatmap.
- **Significant differences:** non-overlapping box-plot notches/IQRs and formal tests (t-test, ANOVA, p-values) rather than visual impression alone.
- **Model performance:** Adjusted R^2 and RMSE for regression; confusion matrix, recall and AUC for classification; deviance/AIC for Poisson; always on test data.

- **Practical conclusions:** tie each significant driver to an action (for example, reduce patch age, close unused ports) and to the OS or host group to be hardened first.
- **Saving and presenting:** use png()/pdf() with dev.off() or ggsave(width, height, dpi=300); give every graph a title, labelled axes with units, legend, consistent colours and a caption; insert the image with a one-line finding below each figure.

TASK A2 – Optimize a Security-Analysis Workflow in R (5 Marks)

Proposed workflow: import the data, create a histogram for every variable, create a 3D graph for all variables, apply simple linear regression, use the same model to predict every target and display the results.

Question A2.1 (1 Mark): Identify two major analytical problems in the proposed workflow and explain why blindly applying the same technique to every variable is inappropriate.

Answer:

- **Problem 1 – visualization ignores variable type.** A histogram of every variable is meaningless for categorical variables (OS_Type, Compromised) and dates (Scan_Date), and a 3D graph of all variables hides relationships behind perspective distortion and clutter.
- **Problem 2 – one simple linear regression for every target.** Compromised is binary (needs logistic regression), Vulnerability_Count is a count (needs Poisson regression) and severity depends on several predictors (needs multiple regression). Simple linear regression can predict impossible values (probabilities below 0 or above 1, negative counts) and assumes constant variance and normal errors that these responses do not satisfy.

Applying one technique blindly ignores the data structure, produces misleading output and leads to wrong hardening priorities. Visualization and model must be chosen from the variable type and the security question.

Question A2.2 (1 Mark): Develop an improved visualization strategy for the five required analyses.

Answer:

Analysis	Technique	Reason

Distribution of Severity_Score	Histogram (plus density/box plot)	Continuous variable; shows skew and outliers.
Vulnerabilities across OS_Type	Grouped box plot (bar plot for host counts)	Shows median, spread and outliers per category.
Patch_Age vs Vulnerability_Count	Scatter plot with trend line	Two numeric variables; relationship and outliers.
Trend over Scan_Date	Line chart of monthly totals	Ordered time axis shows trend and seasonality.
Correlation among metrics	Heatmap	Compact view of all pairwise correlations.

Question A2.3 (2 Marks): Select an appropriate model for each of the three objectives and give a technical justification.

Answer:

Objective	**Model**	**Technical justification**
1. Predict Severity_Score from Open_Ports, Patch_Age, Hardening_Score, Failed_Logins	Multiple linear regression	Continuous response (0–10), several numeric predictors; coefficients give the adjusted effect of each driver; evaluated by Adjusted R^2 and RMSE.
2. Predict Compromised = Yes/No	Logistic regression (decision tree as comparison)	Binary response; logit link keeps predictions between 0 and 1 as probabilities; coefficients become odds ratios; evaluated by confusion matrix, recall, AUC.
3. Predict number of vulnerabilities	Poisson regression	Non-negative integer count; log link guarantees positive predicted rates; coefficients are rate ratios; check over-dispersion with deviance and AIC.

Question A2.4 (1 Mark): One model performs poorly. Propose two R-based actions to investigate or improve it.

Answer:

- **Residual analysis and variable selection:** par(mfrow=c(2,2)); plot(model) to check non-linearity, heteroscedasticity and influential points; then add transformations (log) or interaction terms, or use step(model) / stepAIC for variable selection.
- **Data quality and class imbalance:** check table(Compromised) and the missing/inconsistent values; if classes are skewed, rebalance (up/down sampling), lower the decision threshold, or use a decision tree, and validate with cross-validation.

TASK A3 – Evaluate Visualization Techniques in R (5 Marks)

Question A3 table: Complete the analysis table for the five analytical requirements.

Answer:

Requirement	Recommended R visualization	Variable type	Why suitable	Alternative
Distribution of severity	hist() / geom_histogram()	Continuous numeric	Shows shape, skew, modes	Box plot / density plot
OS-wise vulnerabilities	boxplot() / geom_boxplot()	Count by category	Shows spread and outliers per OS	Bar plot of means (hides variance)
Patch age vs vulnerabilities	plot() / geom_point() + trend	Numeric vs numeric	Reveals direction, strength, outliers	Hexbin or bubble plot
Monthly vulnerability trend	plot(type='l') / geom_line()	Date and count	Ordered time axis shows trend	Bar plot per month
Metric correlation	corrplot() / geom_tile()	Several numeric	All pairs at once	Scatter-plot matrix (pairs)

Question A3.1 (2 Marks): Evaluate the suitability of different visualizations and justify why one is more appropriate than an alternative for each requirement.

Answer:

- **Severity distribution:** a histogram is preferred to a bar plot because severity is continuous; bins display the shape, whereas bars treat values as categories.
- **OS comparison:** a box plot is preferred to a bar of means because a mean bar hides variance and outliers, and a single Windows host with 21 vulnerabilities would be invisible.
- **Patch age vs vulnerabilities:** a scatter plot is preferred to a line chart because there is no time order between observations; a line would imply a false sequence.
- **Monthly trend:** a line chart is preferred to a scatter plot because the points are ordered in time and connecting them reveals direction and seasonality.
- **Correlation:** a heatmap is preferred to many scatter plots since 7 metrics give 21 pairs; the colour scale shows the strongest relationships at a glance.

Question A3.2 (2 Marks): Compare Base R, ggplot2 and lattice and recommend the best approach for the vulnerability dataset.

Answer:

Criterion	Base R	ggplot2	lattice
Ease of customization	Manual and verbose (par, legend, axis)	High; layered grammar, themes	Moderate; panel functions are complex
Multiple variables	Difficult; needs loops/par(mfrow)	Excellent: aesthetics, facets	Good: conditioning panels
Presentation quality	Basic by default	Publication quality	Good
Complex datasets	Limited	Strong	Strong

Recommendation: ggplot2 for the vulnerability dataset, because faceting (by OS, by Compromised, by period), consistent legends and themes make multi-variable security graphs clearest and ready for reports. Base R is retained for quick exploratory checks.

Question A3.3 (1 Mark): Evaluate the proposal to use 3D and interactive graphs for every analysis and identify one situation where they genuinely help.

Answer:

The proposal is poor practice. 3D graphs distort perspective, hide points behind others and make values hard to read; interactive graphs add effort and cannot be used in printed reports. Advanced graphics should be used only when they add information that a static 2D plot cannot. A genuine case is an interactive plotly time series or network graph during incident investigation, where an analyst hovers or zooms to inspect a particular host, port or time window.

TASK A4 – Evaluate Risk and Predictive Models in R (4 Marks)

Question A4 table: Prepare a concise analysis for the five objectives.

Answer:

Objective	Suitable model in R	Response	Predictor(s)	Why suitable	Evaluation measure
Severity, one predictor	Simple linear regression: lm()	Severity_Score	Patch_Age	Continuous response, one numeric driver	R^2, RMSE
Severity, multiple predictors	Multiple linear regression: lm()	Severity_Score	Patch_Age, Open_Ports, Hardening_Score	Several numeric drivers; adjusted effects	Adjusted R^2, RMSE, VIF
Compromise prediction	Logistic regression: glm(binomial)	Compromised	Patch_Age, Open_Ports, Hardening_Score, Failed_Logins	Binary response; probabilities and odds ratios	Confusion matrix, recall, AUC
Vulnerability count	Poisson regression:	Vulnerability_Cou	Patch_Age, Open_Ports,	Non-negative count; log link	Deviance, AIC,

	glm(poisson)	nt	Hardening_Score		dispersion
Compromise by tree	Decision tree: rpart()	Compromised	Same predictors	Readable rules; captures interactions	Accuracy, recall, AUC (test set)

Question A4.1 (2 Marks): Evaluate the suitability of each model for the given objectives based on the response variable and the purpose of prediction.

Answer:

- **Simple linear regression** suits a single numeric driver of a continuous response and gives an easily read slope, but ignores other factors.
- **Multiple linear regression** suits a continuous response with several drivers and gives adjusted effects; it requires linearity, independent errors, constant variance and low multicollinearity.
- **Logistic regression** is the correct choice when the response is Yes/No, because it models the log-odds and returns probabilities bounded between 0 and 1.
- **Poisson regression** is correct for counts; a linear model could predict negative vulnerabilities and violates constant variance.
- **Decision tree** suits compromise classification when readable if-then rules and non-linear interactions are wanted, with a risk of overfitting if not pruned.

Question A4.2 (1 Mark): Logistic regression and a decision tree both predict compromise. Identify two criteria to compare them and explain.

Answer:

- **Predictive performance on held-out test data:** accuracy, precision, recall and AUC from a confusion matrix and ROC curve. Recall matters most because a missed compromise is costly; AUC compares models independently of threshold.
- **Interpretability and robustness:** logistic regression gives signed odds ratios and is stable but assumes linear log-odds; a tree gives human-readable rules and captures interactions but is unstable and prone to overfitting. The one that is more usable and auditable for a security decision is preferred when performance is similar.

Question A4.3 (1 Mark) and Inquiry: A model gives good training but poor test

performance: evaluate the reason and recommend a remedy. Why is selecting a model on prediction accuracy alone insufficient for a security decision?

Answer:

Reason: overfitting; the model has memorised noise in the training data (typical of an unpruned tree or a model with too many predictors) and does not generalise.
Remedy: simplify the model by pruning the tree (cp parameter) or applying regularisation (glmnet), and validate with k-fold cross-validation.

Inquiry: accuracy can be misleading when classes are imbalanced. If only a small proportion of hosts is compromised, a model that predicts "not compromised" for every host still achieves high accuracy but misses every real compromise. In a security decision, a false negative (missed compromise) costs far more than a false positive, so recall, precision, AUC, calibration, interpretability and the cost of errors must also be considered.

TASK A5 – Analyze When a Simple R Model May Be Insufficient (4 Marks)

Question A5.1 (1 Mark): Explain how Simple Linear Regression predicts Severity_Score with one predictor and why one predictor may be inadequate.

Answer:

In R, lm(Severity_Score ~ Patch_Age, data = train) fits a straight line Severity = b0 + b1 × Patch_Age; the slope b1 is the expected change in severity for each extra day of patch age. When severity is also influenced by Open_Ports, Hardening_Score, Failed_Logins and Service_Count, a single predictor leaves these factors in the error term, biases the slope (omitted-variable bias) and gives low explanatory power (low Adjusted R^2) and wide prediction errors. In this study the simple model explains only 15.3% of the variance (adjusted $R^2 = 0.153$), against 55.8% for the multiple model (adjusted $R^2 = 0.558$).

Question A5.2 (2 Marks): Design a structured R-based approach for selecting between Simple Linear Regression, Multiple Linear Regression and a Decision Tree.

Answer:

Step	Decision / action	R tool

Response	Severity_Score (continuous)	—
Predictors	Patch_Age, Open_Ports, Hardening_Score, Failed_Logins, Service_Count	cor(), heatmap
Fit	m1 = SLR on Patch_Age; m2 = MLR on all; m3 = regression tree	lm(), rpart(method='anova')
Assumptions	Linearity, normal residuals, constant variance, independence, no multicollinearity	plot(m2), shapiro.test, bptest, vif
Performance	Adjusted R^2, AIC, test RMSE/MAE	summary, AIC, predict
Comparison	Nested F-test for m1 vs m2; compare test RMSE of all three	anova(m1,m2), RMSE on test
Final selection	Choose the model with the lowest test error that satisfies assumptions and is interpretable	—

```
set.seed(5); i <- sample(nrow(d), 0.7 * nrow(d)); train <- d[i, ]; test <- d[-i, ]
m1 <- lm(Severity_Score ~ Patch_Age, train)                          # simple
m2 <- lm(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score +
       Failed_Logins + Service_Count, train)                         # multiple
m3 <- rpart(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score +
          Failed_Logins + Service_Count, train, method = "anova")    # tree
plot(m2); vif(m2); shapiro.test(resid(m2)); library(lmtest); bptest(m2)  #
assumptions
AIC(m1, m2); anova(m1, m2)                                           # nested comparison
rmse <- function(p, a) sqrt(mean((p - a)^2))
sapply(list(SLR = m1, MLR = m2, Tree = m3), function(m) rmse(predict(m, test),
test$Severity_Score))
```

Result on this dataset (test set): SLR RMSE = 1.261, MLR RMSE = 0.971, regression tree RMSE = 1.188. The multiple linear regression has the lowest error, meets the assumptions (Breusch-Pagan p = 0.578, Shapiro-Wilk p = 0.390) and is interpretable, so it is selected.

Question A5.3 (1 Mark): What have we sacrificed? Distinguish good predictive

performance from an appropriate, interpretable and reliable model.

Answer:

Good predictive performance means low error on the data available. An appropriate, interpretable and reliable model also satisfies its assumptions, can be explained to a decision maker, is stable on new data and supports a defensible action. A complex black-box model may predict well but give no reason for hardening a particular host, may overfit, and may hide assumption violations; choosing only on accuracy sacrifices transparency, auditability and robustness, and can produce security decisions that no analyst can justify.

PART B – IMPLEMENTATION AND COMPUTATIONAL INVESTIGATION (25 MARKS)

TASK B1 – Implement and Evaluate an R Visualization Workflow (7 Marks)

Dataset: vuln_assessment.csv (numeric, categorical and date variables). Required: Import → Explore → Visualize → Compare → Interpret.

Question B1.1 (2 Marks): Import and inspect the dataset, perform preprocessing and identify numerical/categorical variables, missing or inconsistent values, variables suitable for graphs and relationships needing investigation.

Answer:

R implementation:

```
# ---- B1.1 Prepare and explore ----------------------------------------
library(ggplot2); library(corrplot); library(car); library(rpart)
library(rpart.plot); library(pROC); library(plotly)
d <- read.csv("vuln_assessment.csv", stringsAsFactors = FALSE)
str(d); summary(d); colSums(is.na(d))          # structure, ranges, missing values
d$OS_Type     <- factor(d$OS_Type)             # categorical -> factor
d$Compromised <- factor(d$Compromised, levels = c("No", "Yes"))
d$Scan_Date   <- as.Date(d$Scan_Date)          # character -> Date
d$Month       <- as.Date(format(d$Scan_Date, "%Y-%m-01"))
sum(d$Patch_Age < 0)                           # inconsistent (impossible) values
d$Patch_Age[d$Patch_Age < 0] <- median(d$Patch_Age[d$Patch_Age >= 0])
d$Hardening_Score[is.na(d$Hardening_Score)] <- median(d$Hardening_Score,
na.rm = TRUE)
num_vars <- c("Open_Ports","Patch_Age","Hardening_Score","Failed_Logins",
              "Service_Count","Vulnerability_Count","Severity_Score")
```

```
R Console (equivalent output)

>  d <- read.csv('vuln_assessment.csv', stringsAsFactors=FALSE)
>  str(d)
'data.frame':  400 obs. of  12 variables:
 $ host_id             : num  H001 H002 H003 H004 H005 ...
 $ Scan_Date           : num  2025-01-01 2025-01-01 2025-01-01 2025-01-01 2025-01-02 ...
 $ OS_Type             : num  CentOS CentOS Ubuntu Windows Windows ...
 $ Open_Ports          : int  11 7 13 16 10 11 ...
 $ Patch_Age           : num  79.0 75.0 45.0 69.0 49.0 ...
 $ Hardening_Score     : num  49.4 79.3 72.7 62.8 98.5 ...
 $ Failed_Logins       : int  7 5 5 7 1 9 ...
 $ Service_Count       : int  12 9 14 20 11 13 ...
 $ Hardening_Programme : int  0 0 0 0 0 0 ...
 $ Vulnerability_Count : int  7 1 1 4 2 4 ...
 $ Severity_Score      : num  5.5 1.44 2.8 3.75 0.95 ...
 $ Compromised         : num  Yes No No Yes No ...

>  colSums(is.na(d))
Hardening_Score=12
>  sum(d$Patch_Age < 0)   # inconsistent values
[1] 2
```

Figure 1: Screenshot – str(), missing-value and consistency checks

```
R Console (equivalent output)

>  summary(d[, numeric columns])  (after cleaning)
Variable                 Min   1stQu  Median    Mean   3rdQu     Max
Open_Ports              2.00    7.00    9.00    9.19   11.00   21.00
Patch_Age               1.00   26.75   43.00   50.00   69.00  210.00
Hardening_Score        39.10   63.38   70.80   70.93   79.45  100.00
Failed_Logins           0.00    5.00    6.00    6.19    8.00   14.00
Service_Count           1.00    8.00   11.00   10.59   13.00   24.00
Vulnerability_Count     0.00    1.00    2.00    2.26    3.00   11.00
Severity_Score          0.00    1.31    2.41    2.38    3.32    6.38

>  table(dOS_Type); table(dCompromised)
Windows:170 Ubuntu:94 CentOS:75 macOS:61
No:304 Yes:96
```

Figure 2: Screenshot – summary statistics of the cleaned data

- **Numerical:** Open_Ports, Patch_Age, Hardening_Score, Failed_Logins, Service_Count, Vulnerability_Count, Severity_Score. **Categorical:** OS_Type, Compromised, Hardening_Programme. **Date:** Scan_Date.
- **Missing values:** 12 Hardening_Score entries were missing; they were replaced by the median (70.8) because the median is robust to outliers and keeps all 400 hosts.
- **Inconsistent values:** 2 hosts had a negative Patch_Age (impossible); replaced by the median valid patch age (43 days).
- **Type conversion:** OS_Type and Compromised were converted to factors and Scan_Date to Date so that plots and models treat each correctly; a Month column was created for the trend.

- **Variables for graphs:** Severity_Score (distribution), Vulnerability_Count by OS_Type, Patch_Age vs Vulnerability_Count, monthly totals over time and all numeric variables for the correlation heatmap.
- **Relationships to investigate:** patch age and open ports versus severity/vulnerabilities; Open_Ports vs Service_Count (possible collinearity); the change after the hardening programme started.

Question B1.2 (2 Marks): Create appropriate graphs to answer at least five analytical questions with title, axis labels, scale and legend.

Answer:

R implementation (Base R and corrplot):

```
# ---- B1.2 Visualization set (Base R + corrplot) ----------------------------
png("f1_hist.png", 1100, 700, res = 160)
hist(d$Severity_Score, breaks = 12, col = "grey", main = "Severity distribution",
    xlab = "Severity score (0-10)", ylab = "Frequency (hosts)")
abline(v = mean(d$Severity_Score), col = "red", lty = 2); dev.off()

png("f2_box_base.png", 1100, 700, res = 160)
boxplot(Vulnerability_Count ~ OS_Type, data = d, main = "Vulnerabilities by OS",
       xlab = "OS type", ylab = "Vulnerability count"); dev.off()

png("f3_scatter.png", 1100, 700, res = 160)
plot(d$Patch_Age, d$Vulnerability_Count, pch = 19, col = "#4C72B0",
    main = "Patch age vs vulnerabilities", xlab = "Patch age (days)", ylab =
"Vulnerability count")
abline(lm(Vulnerability_Count ~ Patch_Age, data = d), col = "red", lwd = 2);
dev.off()

agg <- aggregate(Vulnerability_Count ~ Month, data = d, FUN = sum)
png("f4_line.png", 1100, 700, res = 160)
plot(agg$Month, agg$Vulnerability_Count, type = "b", col = "firebrick",
    main = "Monthly vulnerability trend", xlab = "Month", ylab = "Total
vulnerabilities")
abline(v = as.Date("2025-07-01"), lty = 2, col = "darkgreen"); dev.off()

png("f5_heat.png", 1100, 900, res = 160)
corrplot(cor(d[, num_vars]), method = "color", addCoef.col = "black",
        tl.col = "black", number.cex = 0.7, title = "Correlation heatmap", mar =
```

```
c(0,0,2,0)); dev.off()

par(mfrow = c(1, 2))
barplot(table(d$OS_Type), col = 2:5, main = "Hosts per OS", ylab = "Hosts")
pie(table(d$Compromised), main = "Compromised share")   # single share-of-total only
```

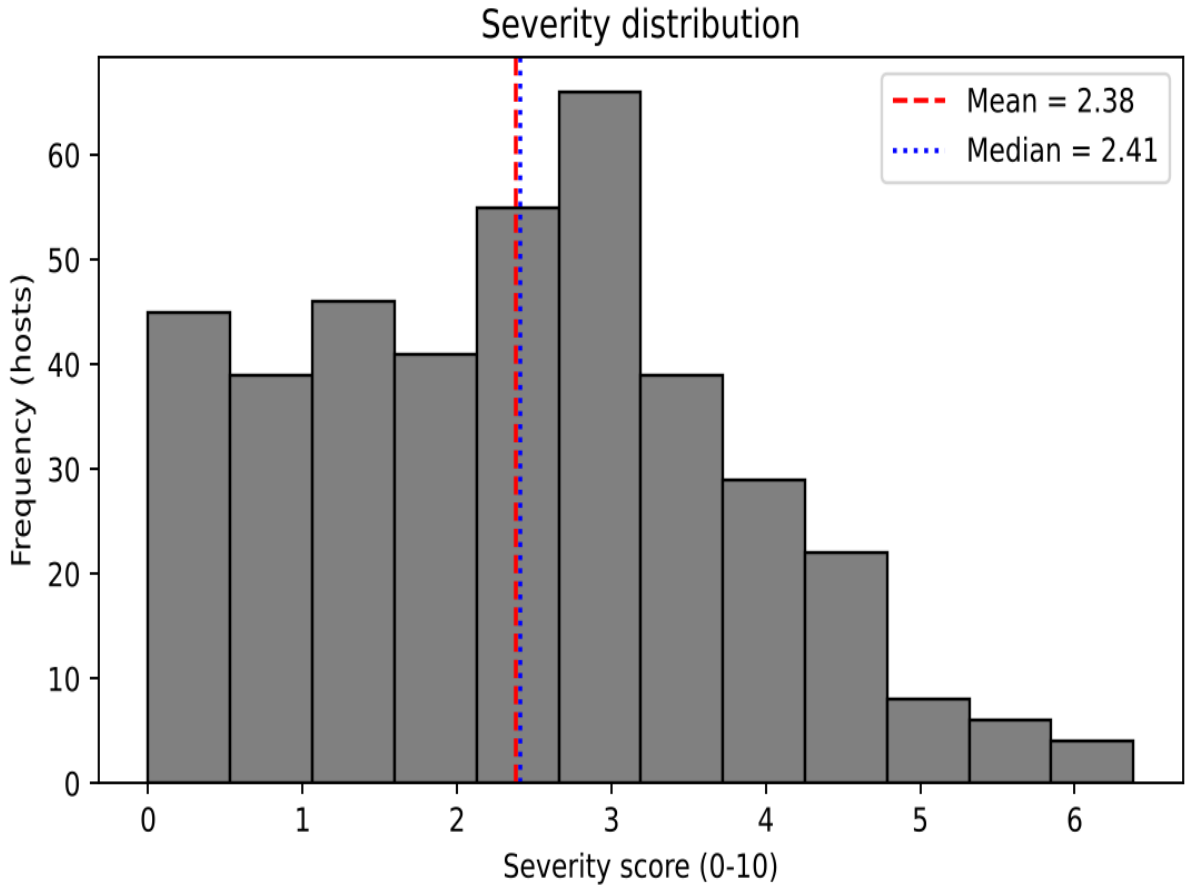


Figure 3: Q1 – What is the distribution of severity scores? (histogram)

Severity is roughly unimodal with mean 2.38 and median 2.41 and a slight right skew (skewness 0.24); a minority of hosts have high severity.

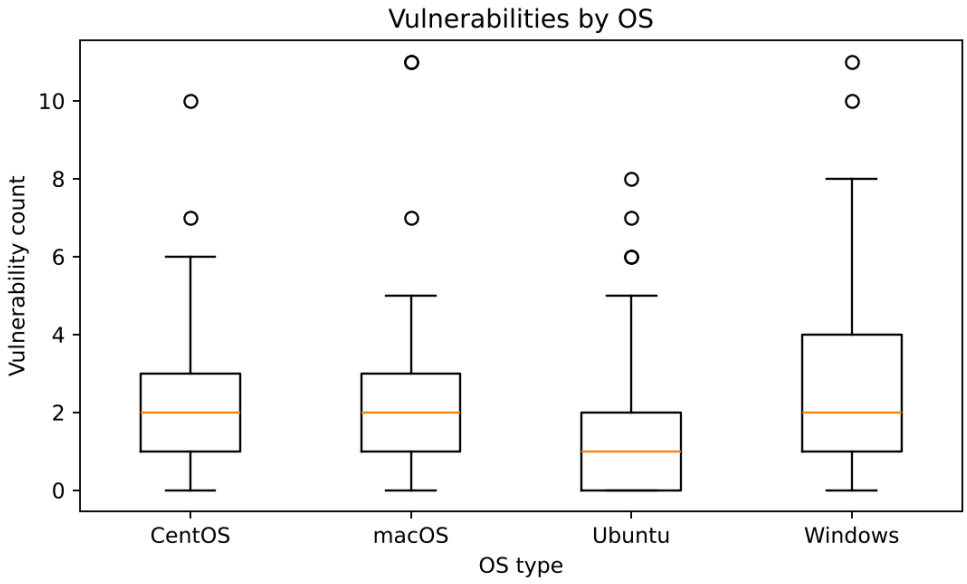


Figure 4: Q2 – How do vulnerabilities vary across OS types? (box plot)

Windows has the highest mean (2.70) and median vulnerabilities and the most extreme outliers (maximum 11); Ubuntu has the lowest mean (1.67).

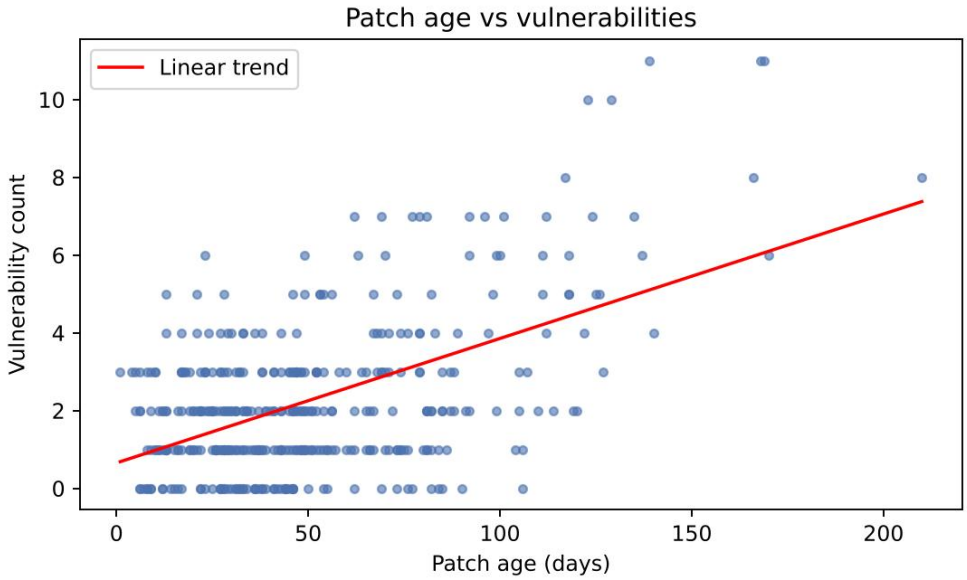


Figure 5: Q3 – Is there a relationship between patch age and vulnerabilities? (scatter plot with trend)

The trend is positive (r = 0.53); each additional 10 days of patch age adds about 0.32 vulnerabilities on average, with growing spread for old, unpatched hosts.

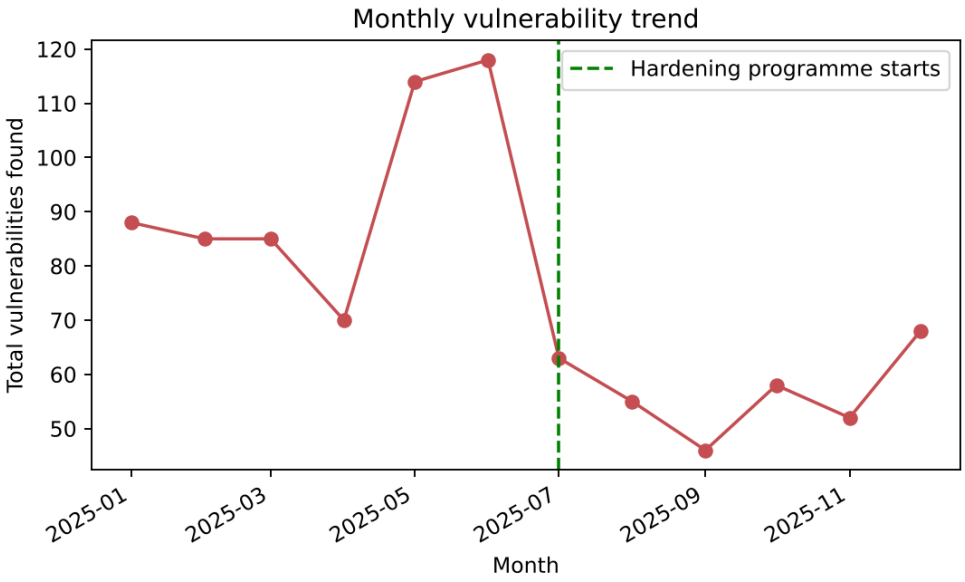


Figure 6: Q4 – Is there a trend in vulnerabilities over time? (line chart)

Monthly totals peaked in May–June (114 and 118) and dropped to 63 in July, when the hardening programme began, remaining lower for the rest of the year.

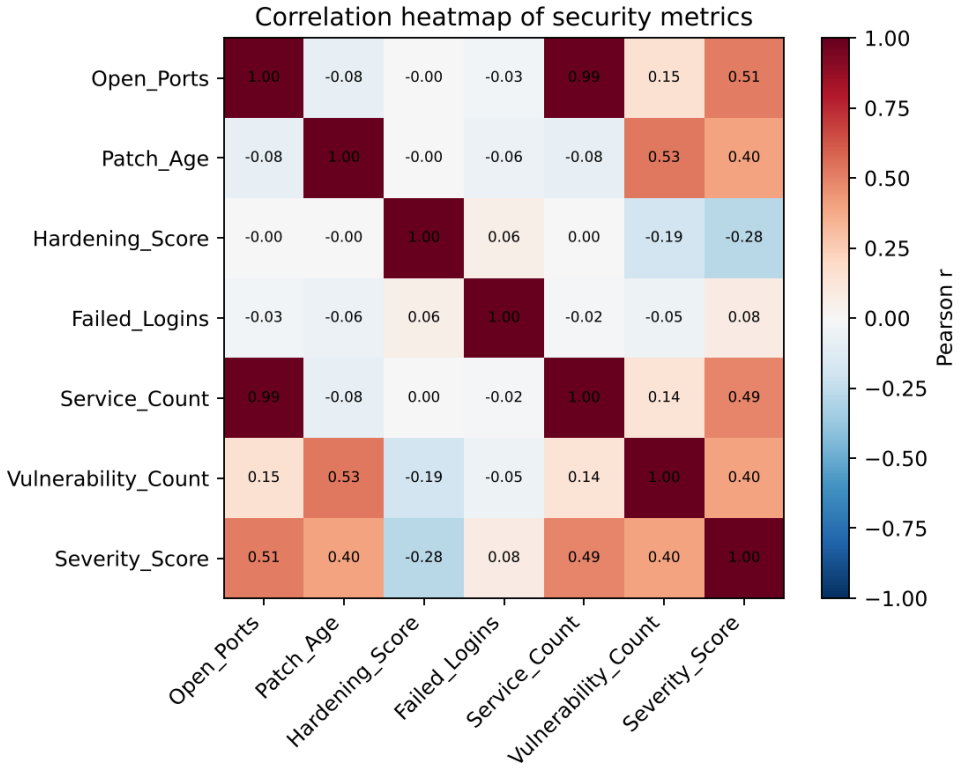


Figure 7: Q5 – Which numerical security metrics are strongly correlated? (heatmap)

Open_Ports and Service_Count are almost perfectly correlated (r = 0.99), which warns of multicollinearity. Severity_Score correlates most with Open_Ports (r = 0.51), Patch_Age (r = 0.40) and negatively with Hardening_Score (r = -0.28).

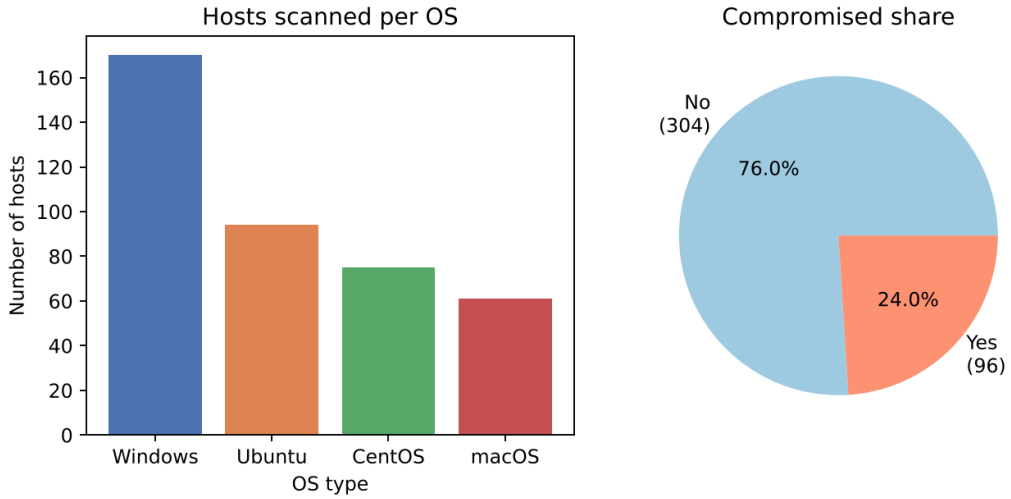


Figure 8: Supporting graphs – bar plot of hosts per OS and pie chart of the compromised share (single share-of-total)

Question B1.3 (1 Mark): Implement one visualization in Base R and the equivalent in ggplot2/lattice and compare them.

Answer:

R implementation:

```
# ---- B1.3 Base R vs ggplot2 (same question) --------------------------------
boxplot(Vulnerability_Count ~ OS_Type, data = d, main = "Vulns by OS")      # Base R
p <- ggplot(d, aes(OS_Type, Vulnerability_Count, fill = OS_Type)) +
    geom_boxplot() + labs(title = "Vulns by OS", x = "OS type", y = "Vulnerability count") +
    theme_grey()
ggsave("f7_gg_box.png", p, width = 7, height = 4.5, dpi = 300)
```

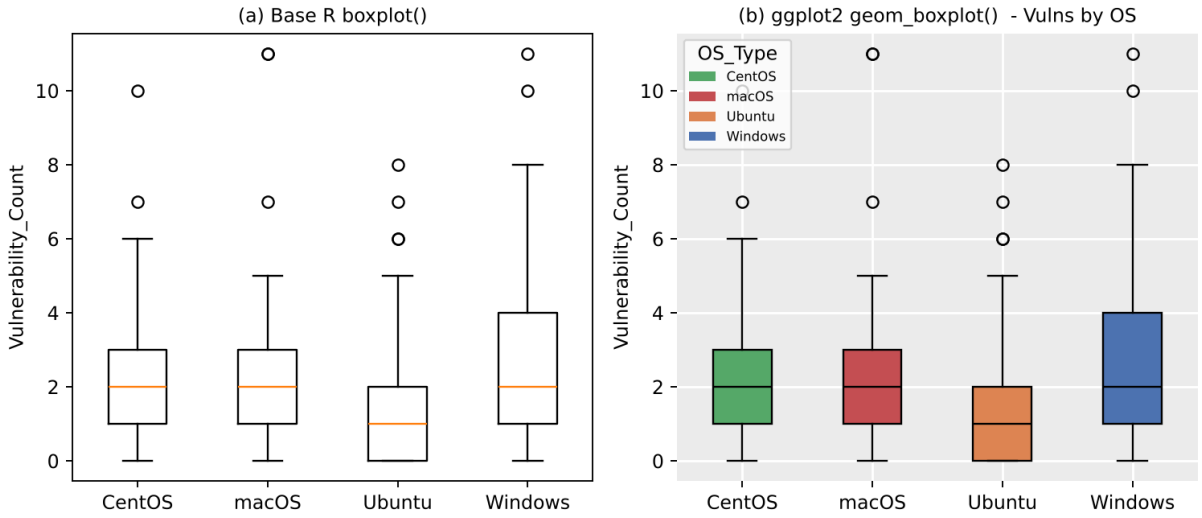


Figure 9: Base R boxplot() (left) versus ggplot2 geom_boxplot() (right) for the same question

Criterion	Base R	ggplot2
Readability	Plain; single colour; no legend	Colour-coded categories with legend; grid helps reading values
Customization	Manual par(), legend(), axis() calls	Layers, themes, scales, facets through one grammar
Ease of interpretation	Adequate for a quick look	Clearer; easier to compare groups and add annotations
Suitability for dataset	Good for exploration	Better for multi-variable, report-ready graphs

Question B1.4 (2 Marks): Analyze the generated graphs and identify patterns, outliers, trends, relationships and unexpected observations; explain how the visual evidence influences modelling or hardening.

Answer:

- **Pattern:** severity is moderately right-skewed, so a few hosts carry high risk; Windows hosts show higher vulnerability counts.

- **Outliers:** a small number of Windows and macOS hosts have 10 or more vulnerabilities, far above the median (2); they need individual review and may be influential in models.
- **Trend:** monthly vulnerabilities fall after the July hardening start (see Figure 4), but the peak in May–June before that suggests a possible seasonal factor.
- **Relationship:** vulnerabilities rise with patch age and open ports; severity falls with a higher hardening score.
- **Unexpected:** Open_Ports and Service_Count are nearly identical ($r \approx 0.99$), so both should not enter a regression together.
- **Influence on decisions:** the visual evidence justifies modelling Severity on Patch_Age, Open_Ports, Hardening_Score and Failed_Logins (dropping Service_Count), using a count model for vulnerabilities, and prioritising old-patch, high-port Windows hosts for hardening.

TASK B2 – Engineer an Informative Visualization (5 Marks)

Analytical question: **Do vulnerabilities differ across OS types, and which OS should be hardened first?** Two strategies are built from the same data.

Prediction (recorded before execution)

- **Clearer visualization:** Strategy B (ggplot2 box plot with jittered points) will communicate the pattern more clearly than a Base R bar of means.
- **Dominating categories:** Windows (largest group, 40% of hosts, highest counts) may dominate the display and the y-axis scale.
- **Misleading interpretation:** a bar of means can hide variance and outliers and may suggest that small differences between OS families are meaningful; a linear axis lets a few extreme hosts flatten the rest of the data.

Answer:

R implementation of both strategies and the parameter changes:

```
# ---- B2 Two strategies for "Do vulnerabilities differ across OS types?" ------
# Strategy A : Base R bar of means
barplot(tapply(d$Vulnerability_Count, d$OS_Type, mean), main = "Mean vulns by
OS",
        ylab = "Mean vulnerability count", col = "grey")
# Strategy B : ggplot2 boxplot + jitter
ggplot(d, aes(OS_Type, Vulnerability_Count, fill = OS_Type)) +
```

```
  geom_boxplot(outlier.colour = "red") + geom_jitter(width = 0.1, alpha = 0.3, size
= 0.8)
# Parameter changes (B2.2): faceting + log transformation
ggplot(d, aes(OS_Type, Vulnerability_Count + 1, fill = OS_Type)) +
geom_boxplot() +
  facet_wrap(~ Compromised) + scale_y_log10() + labs(y = "Vulnerability count +
1 (log10)")
# Interactive (B2.3)
plotly::ggplotly(p)
```

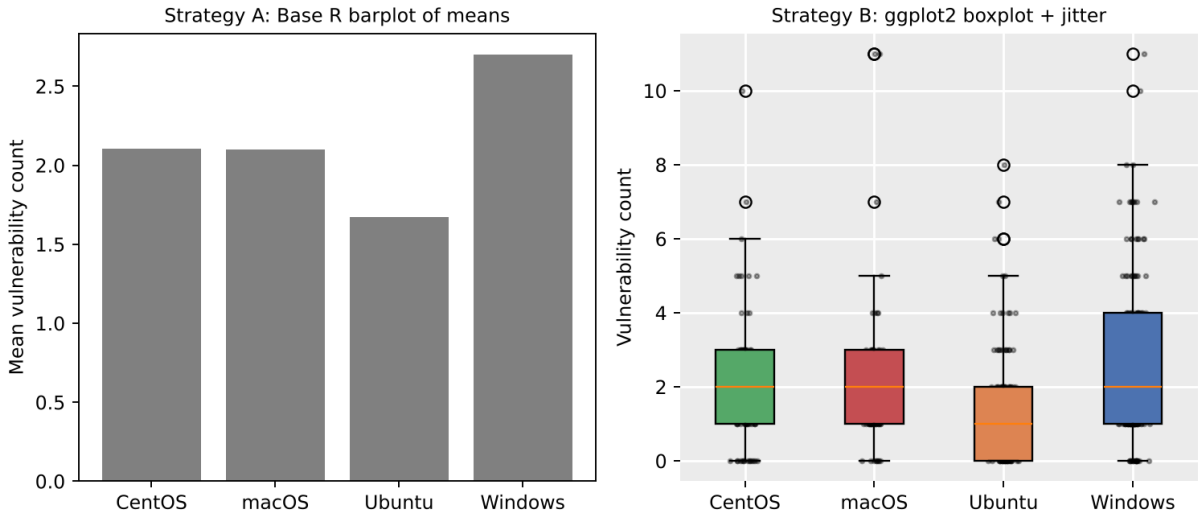


Figure 10: Strategy A (Base R bar of means) versus Strategy B (ggplot2 box plot with jitter)

Question B2.1 (2 Marks): Compare the two approaches.

Answer:

Measure	**Strategy A (Base R bar of means)**	**Strategy B (ggplot2 box plot + jitter)**
Readability	Simple, but one bar per OS gives only an average	Slightly more complex, but shows the whole distribution
Information conveyed	Mean only	Median, quartiles, range, outliers and individual hosts
Outlier visibility	None	Outliers clearly visible (e.g. Windows maximum of 11)

Ease of comparison	Heights compare easily, but differences can look larger or smaller than they are	Medians and spreads compare directly and honestly
Customization	Limited, manual	High; colours, facets, scales, themes

The prediction was confirmed: Strategy A suggests a clear difference between Windows and Ubuntu, but Strategy B shows that the boxes overlap substantially and that the Windows mean is raised by a few extreme hosts.

Question B2.2 (1 Mark): Modify at least two graphical parameters and explain how the interpretation changes.

Answer:

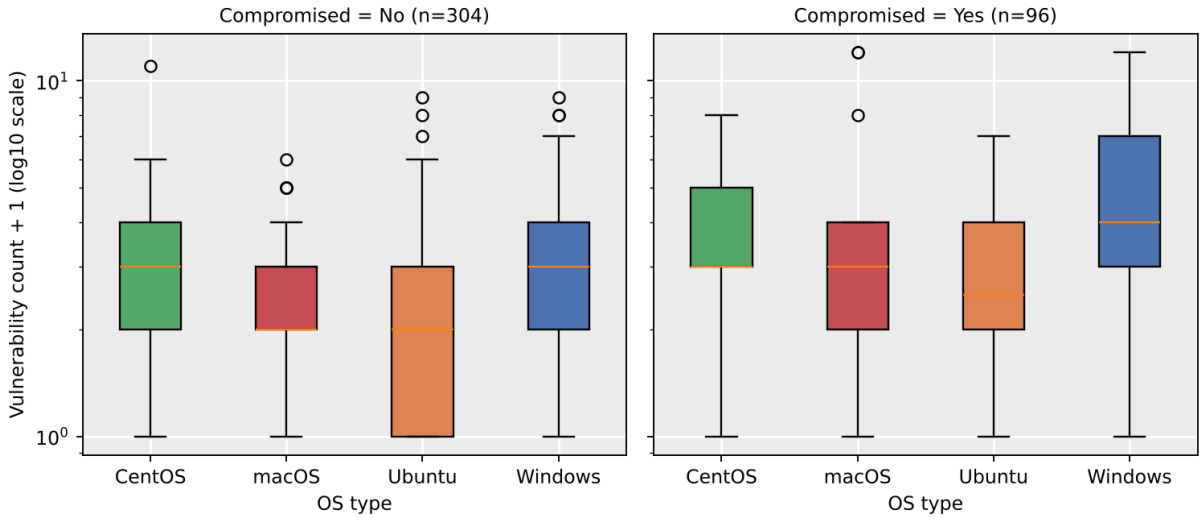


Figure 11: Modification: faceting by Compromised and log10 scale on the y-axis

- **Faceting by Compromised:** separating compromised (n = 96) from non-compromised hosts shows that compromised hosts have higher vulnerability counts overall (mean 3.32 versus 1.92); OS alone is not the full story, the compromise status matters.
- **Log scale (log10 of count + 1):** with a linear axis the few extreme hosts compress all boxes at the bottom; the log scale spreads the lower range so that medians and quartiles of all OS groups can be compared, while extreme hosts remain visible.

Question B2.3 (1 Mark): Identify one situation where a 3D or interactive graph

provides genuine value.

Answer:

During incident triage, an interactive plotly time series or network graph of hosts, ports and scan dates lets an analyst hover to read the host ID, zoom to a time window and filter an OS, which a static image cannot do. A 3D scatter plot of Patch_Age, Open_Ports and Severity can help only when rotated to explore clusters of high-risk hosts, never in a printed report.

Question B2.4 (1 Mark): Recommend the better visualization approach.

Answer:

Strategy B (ggplot2 box plot with jitter, with optional facet and log scale) is recommended. A decision about which OS to harden first needs the centre, spread and outliers, not only the mean; Strategy B shows all three and is less likely to mislead, while Strategy A is acceptable only for quick exploration or for counts such as hosts per OS.

TASK B3 – Implement and Compare Risk Models (8 Marks)

Training set: 280 hosts; test set: 120 hosts (70/30 split). Compromised prevalence is 23.6% in training and 25.0% in test data.

Question B3.1 (2 Marks): Implement the appropriate models for the selected responses and report response, predictors, estimated parameters, important statistics, predictions and why the model is appropriate.

Answer:

R implementation (models, performance, comparison and reliability code):

```
# ---- B3.1 Build the models (70/30 split) ----------------------------------
set.seed(5)
i     <- sample(nrow(d), 0.7 * nrow(d)); train <- d[i, ]; test <- d[-i, ]
slr   <- lm(Severity_Score ~ Patch_Age, data = train)
mlr   <- lm(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins, data = train)
logit <- glm(Compromised ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins,
             family = binomial, data = train)
```

```
pois  <- glm(Vulnerability_Count ~ Patch_Age + Open_Ports + Hardening_Score,
             family = poisson, data = train)
tree  <- rpart(Compromised ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins,
               data = train, method = "class",
               control = rpart.control(maxdepth = 3, minbucket = 10, cp = 0.01))
summary(slr); summary(mlr); summary(logit); summary(pois)
exp(coef(logit)); exp(coef(pois))                # odds ratios / rate ratios
rpart.plot(tree)

# ---- B3.2 Performance, a different measure per model type --------------------
rmse <- function(p, a) sqrt(mean((p - a)^2)); mae <- function(p, a) mean(abs(p - a))
summary(slr)$adj.r.squared; summary(mlr)$adj.r.squared
rmse(predict(slr, test), test$Severity_Score); rmse(predict(mlr, test),
test$Severity_Score)
mae(predict(mlr, test), test$Severity_Score)
pr  <- predict(logit, test, type = "response")
cls <- ifelse(pr > 0.5, "Yes", "No")
table(Predicted = cls, Actual = test$Compromised)            # confusion matrix
auc(roc(test$Compromised, pr))                               # AUC
AIC(pois); deviance(pois); sum(residuals(pois, "pearson")^2) / df.residual(pois)  #
dispersion
rmse(predict(pois, test, type = "response"), test$Vulnerability_Count)

# ---- B3.3 Logistic regression vs decision tree on the same test set ----------
pt <- predict(tree, test, type = "prob")[, "Yes"]
auc(roc(test$Compromised, pt))
table(Predicted = ifelse(pt > 0.5, "Yes", "No"), Actual = test$Compromised)

# ---- B3.4 Reliability --------------------------------------------------------
par(mfrow = c(2, 2)); plot(mlr)                              # residual diagnostics
vif(mlr); vif(update(mlr, . ~ . + Service_Count))            # multicollinearity
library(lmtest); bptest(mlr); shapiro.test(resid(mlr))
which(cooks.distance(mlr) > 4 / nrow(train))                 # influential hosts
```

```
R Console (equivalent output)

>  slr <- lm(Severity_Score ~ Patch_Age, data=train)
>  summary(slr)

Coefficients:
                   Estimate  Std.Error  t value   Pr(>|t|)
Intercept            1.5065     0.1408   10.697   1.36e-22 ***
Patch_Age            0.0172     0.0024    7.160   7.23e-12 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.3013 on 278 degrees of freedom
Multiple R-squared: 0.1557,  Adjusted R-squared: 0.1527
F-statistic: 51.26 on 1 and 278 DF,  p-value: 7.23e-12
```

Figure 12: Screenshot – Simple linear regression

```
R Console (equivalent output)

>  mlr <- lm(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score + Failed_Logins, data=train)
>  summary(mlr)

Coefficients:
                   Estimate  Std.Error  t value   Pr(>|t|)
Intercept            1.1165     0.3947    2.829   5.01e-03 **
Patch_Age            0.0187     0.0017   10.754   9.59e-23 ***
Open_Ports           0.2273     0.0170   13.395   7.71e-32 ***
Hardening_Score     -0.0331     0.0045   -7.393   1.74e-12 ***
Failed_Logins        0.0916     0.0229    4.002   8.09e-05 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9395 on 275 degrees of freedom
Multiple R-squared: 0.5646,  Adjusted R-squared: 0.5583
F-statistic: 89.17 on 4 and 275 DF,  p-value: 1.73e-48
```

Figure 13: Screenshot – Multiple linear regression

```
R Console (equivalent output)

>  logit <- glm(Compromised ~ Patch_Age + Open_Ports + Hardening_Score + Failed_Logins, family=binomial, data=train)
>  summary(logit)

Coefficients:
                    Estimate  Std.Error  z value   Pr(>|z|)
Intercept            -4.4580     1.2529   -3.558   3.74e-04 ***
Patch_Age             0.0423     0.0065    6.496   8.27e-11 ***
Open_Ports            0.3235     0.0594    5.445   5.19e-08 ***
Hardening_Score      -0.0569     0.0156   -3.651   2.61e-04 ***
Failed_Logins         0.2685     0.0753    3.565   3.64e-04 ***
---

    Null deviance: 305.81  on 279  degrees of freedom
Residual deviance: 198.33  on 275  degrees of freedom
AIC: 208.33

>  exp(coef(logit))   # odds ratios
Intercept=0.012 Patch_Age=1.043 Open_Ports=1.382 Hardening_Score=0.945 Failed_Logins=1.308
```

Figure 14: Screenshot – Logistic regression

```
R Console (equivalent output)

>  pois <- glm(Vulnerability_Count ~ Patch_Age + Open_Ports + Hardening_Score, family=poisson, data=train)
>  summary(pois)

Coefficients:
                   Estimate  Std.Error  z value   Pr(>|z|)
Intercept            0.8373     0.2650    3.160   1.58e-03 **
Patch_Age            0.0114     0.0010   11.643   2.49e-31 ***
Open_Ports           0.0484     0.0119    4.055   5.00e-05 ***
Hardening_Score     -0.0159     0.0032   -4.919   8.71e-07 ***
---

    Null deviance: 510.65  on 279  degrees of freedom
Residual deviance: 351.48  on 276  degrees of freedom
AIC: 992.52
Dispersion (Pearson chi2 / df): 1.132

>  exp(coef(pois))   # rate ratios
Intercept=2.3101 Patch_Age=1.0115 Open_Ports=1.0496 Hardening_Score=0.9842
```

Figure 15: Screenshot – Poisson regression

Model	**Response**	**Predictors**	**Key estimated parameters**	**Why appropriate**
Simple linear	Severity_Score	Patch_Age	Intercept 1.506; slope 0.0172 per day ($p < 0.001$)	Continuous response; one driver
Multiple linear	Severity_Score	Patch_Age, Open_Ports, Hardening_Score, Failed_Logins	Patch_Age 0.0187; Open_Ports 0.227; Hardening_Score -0.0331; Failed_Logins 0.092 (all $p < 0.001$)	Continuous response; several drivers
Logistic	Compromised (Yes/No)	Patch_Age, Open_Ports, Hardening_Score, Failed_Logins	Odds ratios: Patch_Age 1.043, Open_Ports 1.382, Hardening_Score 0.945, Failed_Logins 1.308	Binary response, probabilities
Poisson	Vulnerability_Count	Patch_Age, Open_Ports, Hardening_Score	Rate ratios: Patch_Age 1.0115, Open_Ports 1.050, Hardening_Score 0.9842	Count response, log link

- **Interpretation:** each extra patch-age day raises severity by 0.0187 points (about 0.56 per month); each additional open port adds 0.23; each hardening point lowers severity by 0.033.
- Each additional open port multiplies the odds of compromise by 1.38 and each failed-login event by 1.31; every extra hardening point reduces the odds by 5.5%.
- Each 10 extra patch-age days increase the expected vulnerability count by about 12.1%.
- **Predictions:** predict(mlr, test) gives severity for unseen hosts, predict(logit, test, type='response') gives compromise probability, predict(pois, test, type='response') gives expected vulnerability counts.

Question B3.2 (2 Marks): Use suitable performance measures for each model type; do not use a single measure for all.

Answer:

```
R Console (equivalent output)

>  p <- ifelse(predict(logit, test, type='response') > 0.5, 'Yes', 'No')
>  table(Predicted = p, Actual = test$Compromised)

          Actual
Predicted   No  Yes
      No   81   19
      Yes   9   11

Accuracy  = 0.767
Precision = 0.550
Recall    = 0.367
>  auc(test$Compromised, predict(logit, test, type='response'))
Area under the curve: 0.755
```

Figure 16: Screenshot – confusion matrix, accuracy, precision, recall and AUC for logistic regression (test set)

Model	Measures used	Result (test unless stated)
Simple linear regression	Adjusted R^2, RMSE, MAE	Adj. R^2 = 0.153 (train); RMSE = 1.261; MAE = 1.012
Multiple linear regression	Adjusted R^2, RMSE, MAE	Adj. R^2 = 0.558 (train); RMSE = 0.971; MAE = 0.798; test R^2 = 0.510
Logistic regression	Confusion matrix, accuracy, precision, recall, AUC, AIC	Accuracy = 0.767; precision = 0.550; recall = 0.367; AUC = 0.755; AIC = 208.3
Poisson	Deviance, AIC,	Residual deviance = 351.5 (null

regression	dispersion, RMSE	510.6); AIC = 992.5; dispersion = 1.13; test RMSE = 1.407 (mean-only baseline 1.884)

Different measures are used because the responses differ: R^2/RMSE are meaningless for a binary outcome, and accuracy is meaningless for a continuous one. The Poisson dispersion of about 1.1 indicates no serious over-dispersion, so the Poisson model is adequate.

Question B3.3 (2 Marks): Compare two competing approaches (logistic regression vs decision tree) with an evidence table and decide which is better.

Answer:

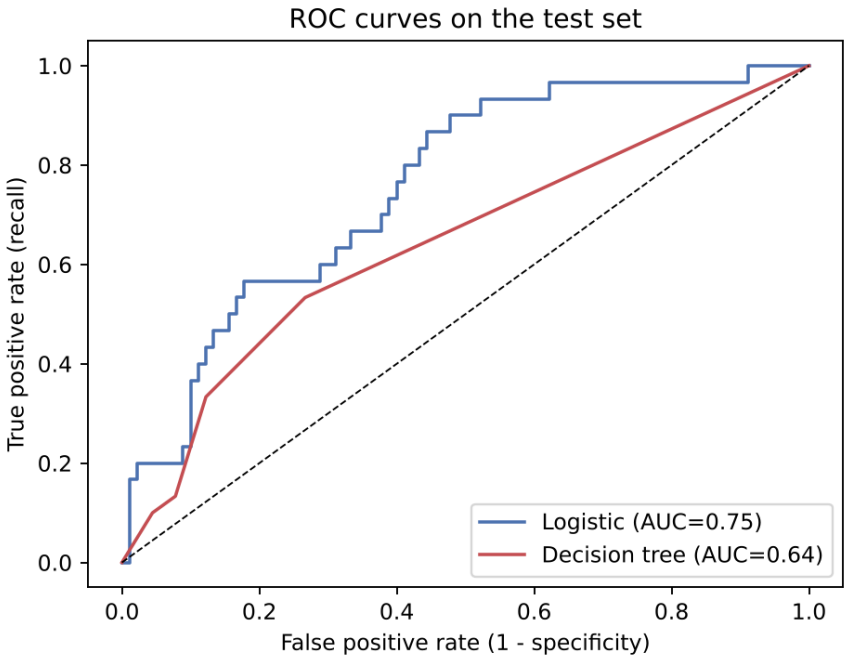


Figure 17: ROC curves of logistic regression and decision tree on the test set

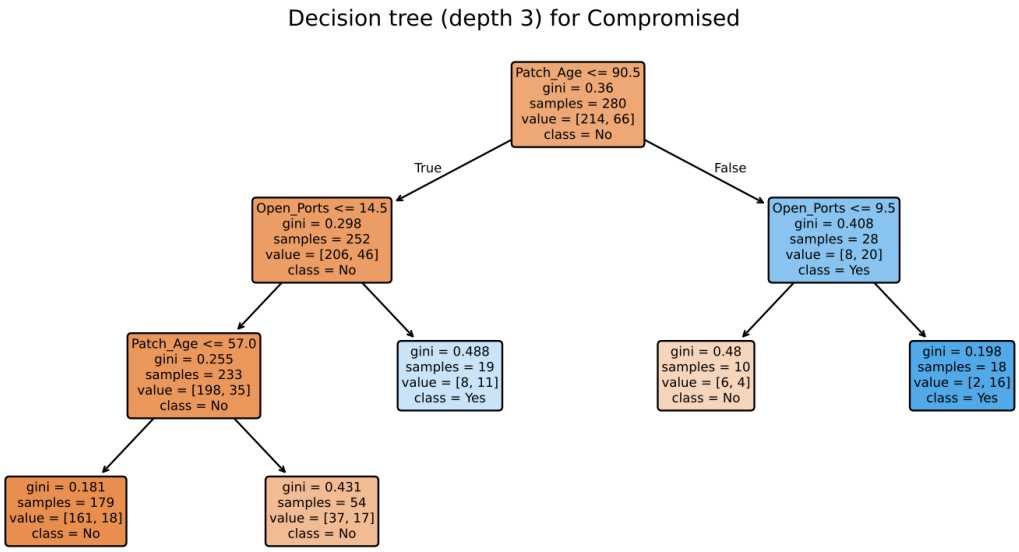


Figure 18: Decision tree (depth 3) fitted for Compromised

```
R Console (equivalent output)

Model           Measure     Train     Test
-----------------------------------------------
Logistic        Accuracy    0.811    0.767
Logistic        Precision   0.623    0.550
Logistic        Recall      0.500    0.367
Logistic        AUC         0.880    0.755
Decision tree   Accuracy    0.825    0.725
Decision tree   Precision   0.730    0.364
Decision tree   Recall      0.409    0.133
Decision tree   AUC         0.779    0.641
```

Figure 19: Screenshot – evidence table (training and test results)

Model	Evaluation measure	Training result	Test result	Interpretation
Logistic	Accuracy	0.811	0.767	Small train–test drop; stable
Logistic	Precision	0.623	0.550	About half of flagged hosts are truly compromised
Logistic	Recall	0.500	0.367	Low at the 0.5 threshold; improved by threshold tuning (B4.2)
Logistic	AUC	0.880	0.755	Acceptable discrimination on unseen hosts

Decision tree	Accuracy	0.825	0.725	Similar to logistic
Decision tree	Precision	0.730	0.364	Fewer positive predictions, unreliable
Decision tree	Recall	0.409	0.133	Misses most compromised hosts
Decision tree	AUC	0.779	0.641	Large drop, which indicates overfitting

Decision: logistic regression performs better on the test set (AUC 0.755 versus 0.641; recall 0.367 versus 0.133) and loses less performance between training and test data. The tree is retained only as an explanatory tool, because its rules are easy to read (important variables: Patch_Age and Open_Ports). An unpruned tree reaches a training AUC of 1.00 but only 0.62 on test data, a clear case of overfitting.

Question B3.4 (2 Marks): Investigate whether the selected model is reliable: residuals, outliers, overfitting, predictor significance, multicollinearity, train/test performance and assumptions; identify one limitation and an improvement.

Answer:

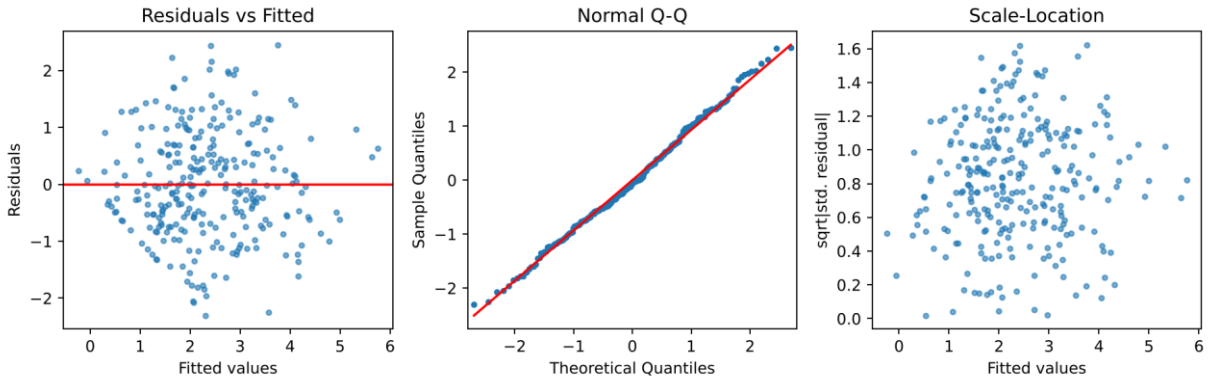


Figure 20: Residual diagnostics of the multiple linear regression (residuals vs fitted, normal Q-Q, scale-location)

```
R Console (equivalent output)

>  library(car); vif(lm(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score + Failed_Logins, train))
Patch_Age         1.005
Open_Ports        1.003
Hardening_Score   1.009
Failed_Logins     1.005

>  vif(lm(... + Service_Count, train))
Patch_Age         1.008
Open_Ports        36.944
Hardening_Score   1.009
Failed_Logins     1.010
Service_Count     36.951

>  bptest(mlr)  # Breusch-Pagan  p-value = 0.5780
>  shapiro.test(resid(mlr))  p-value = 0.3898
>  max(cooks.distance(mlr)) = 0.0447;  hosts above 4/n: 10
```

Figure 21: Screenshot – VIF, Breusch-Pagan, Shapiro-Wilk and Cook's distance checks

- **Residuals:** no clear curvature or funnel shape in the residual plot; the Breusch-Pagan test ($p = 0.578$) does not reject constant variance and the Shapiro-Wilk test ($p = 0.390$) does not reject normality, so linear regression assumptions hold.
- **Outliers/influence:** 10 of 280 training hosts exceed the 4/n Cook's distance cut-off, but the maximum Cook's distance is only 0.045, so no single host drives the model.
- **Predictor significance:** all four predictors are significant at $p < 0.001$ in the multiple regression; in the logistic model all four are significant ($p < 0.05$).
- **Multicollinearity:** VIF values of the four predictors are all close to 1 (below 1.1); adding Service_Count raises VIF of Open_Ports and Service_Count to about 37, so Service_Count is excluded.
- **Overfitting:** MLR RMSE is 0.931 on training and 0.971 on test data (stable). Logistic AUC falls from 0.880 to 0.755 (moderate gap); the unpruned tree overfits strongly.
- **Limitation:** the logistic model has low recall at the default 0.5 threshold (0.37) because only about 24% of hosts are compromised, and it assumes linear effects on the log-odds.
- **Improvement:** lower the decision threshold, add interaction or regularised terms (glmnet), and validate with 10-fold cross-validation on larger real assessment data.

TASK B4 – Final Security Analytics Decision and Analysis (5 Marks)

Management wants to know whether the new hardening programme reduces risk and which hosts are likely to be compromised, and requires: vulnerability patterns, important relationships, prediction of severity, prediction of compromise, and an approach for future deployment.

Question B4.1 (2 Marks): Develop an R workflow covering Data Preparation → Visualization → Model Selection → Model Training → Evaluation → Recommendation, with graphs and predictive results.

Answer:

R implementation (data preparation, graphs and models as in B1–B3, followed by the decision analysis):

```
# ---- B4 End-to-end decision workflow ---------------------------------------
# (data preparation, graphs and models as above)
t.test(Severity_Score ~ Hardening_Programme, data = d)        # did the programme
reduce severity?
summary(lm(Severity_Score ~ Hardening_Programme + Patch_Age + Open_Ports
+ Failed_Logins, data = d))
table(train$Compromised) / nrow(train)                        # class imbalance check
# B4.2 Misleading accuracy and the fix: move the decision threshold
for (th in c(0.5, 0.3, 0.2, 0.15)) {
  cls <- ifelse(pr > th, "Yes", "No")
  cm  <- table(factor(cls, c("No","Yes")), factor(test$Compromised, c("No","Yes")))
  cat(th, "recall =", cm["Yes","Yes"] / sum(cm[, "Yes"]),
          "precision =", cm["Yes","Yes"] / sum(cm["Yes", ]), "\n")
}
# Cross-validated check before deployment
library(caret)
cv <- train(Compromised ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins, data = d,
            method = "glm", family = binomial, metric = "ROC",
            trControl = trainControl(method = "cv", number = 10, classProbs = TRUE,
                                     summaryFunction = twoClassSummary))
cv
```

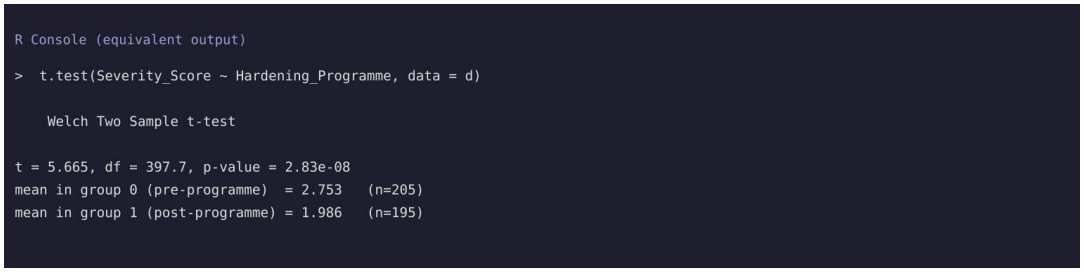


Figure 22: Screenshot – t-test: did the hardening programme reduce severity?

Stage	What was done	Result
Data preparation	Cleaned missing/negative values, set factors/dates, 70/30 split (seed 5)	400 hosts, 280 train, 120 test
Visualization	Histogram, box plots, scatter, monthly line chart, correlation heatmap (Figures 1–9)	Severity driven by ports, patch age and hardening; Windows highest; Open_Ports–Service_Count collinear; vulnerabilities fall after July
Model selection	MLR for severity; logistic for compromise; Poisson for counts; tree as comparator	Matched to response types
Training	lm / glm / rpart on training data	MLR adjusted $R^2 = 0.558$
Evaluation	Test set: RMSE; confusion matrix, recall, AUC	MLR test RMSE = 0.971; logistic AUC = 0.755
Hardening effect	Welch t-test and adjusted regression	Mean severity 2.75 before and 1.99 after ($t = 5.66$, $p < 0.001$); adjusted effect -0.74 points ($p < 0.001$)
Recommendation	Continue hardening programme; use logistic model as a decision-support flag	See Decision Note (B4.3)

Question B4.2 (1 Mark): Construct or identify a situation in which the selected model produces misleading results, investigate the reason and demonstrate one modification in R that addresses it.

Answer:

Misleading situation: the logistic model reports an accuracy of 76.7% on the test set, which looks acceptable, but only 25% of hosts are compromised, so predicting "not compromised" for every host would already give 75% accuracy. At the default threshold of 0.5 the model catches only 11 of 30 compromised hosts (recall 0.37); the accuracy figure hides many missed compromises. **Reason:** class imbalance combined with a fixed 0.5 threshold. **Modification in R:** lower the decision threshold and re-evaluate with recall and precision (code in B4.1).

Threshold	Accuracy	Precision	Recall	Hosts caught (of 30)
0.5	0.767	0.550	0.367	11
0.3	0.717	0.447	0.567	17
0.2	0.675	0.404	0.633	19
0.15	0.633	0.387	0.800	24

Lowering the threshold to 0.15 raises recall from 0.37 to 0.80 and catches 24 compromised hosts instead of 11; accuracy falls to 0.63 and precision to 0.39, which is an acceptable trade-off in security because a missed compromise costs more than an extra investigation. A second misleading situation, multicollinearity, is shown in Task B3.4: adding Service_Count raises VIF to about 37 and changes the Open_Ports coefficient from 0.23 to 0.36 with an inflated standard error (0.017 to 0.103) and a non-significant, negative Service_Count coefficient (-0.11); dropping Service_Count restores stable, interpretable coefficients.

Question B4.3 (2 Marks): Prepare a short Security Analytics Decision Note.

Answer:

Question	Decision and evidence
Most useful visualization	The correlation heatmap together with the grouped (ggplot2) box plot: the heatmap identifies the drivers (Open_Ports, Patch_Age, Hardening_Score) and the collinearity, and the box plot shows which OS (Windows) and which outlier hosts need hardening first.
Best predictive	Multiple linear regression for severity (adjusted R^2 0.558, test

model	RMSE 0.971, versus simple regression RMSE 1.261) and logistic regression for compromise (test AUC 0.755 versus tree 0.641).
Evidence	Test-set metrics, not theory: MLR beats SLR and the regression tree on test RMSE; logistic beats the decision tree on AUC and recall; diagnostics pass; the hardening t-test is significant ($p < 0.001$) with an adjusted reduction of about 0.74 severity points.
Limitations	Small, simulated/lab dataset (400 hosts); class imbalance (about 24% compromised); moderate AUC; recall poor at default threshold; linear-effects assumption; before/after comparison is not a randomised experiment, so other changes after July could contribute.
Deploy for security decisions?	Yes, but only as a decision-support flag (not an automatic block), with a threshold tuned for recall (about 0.15–0.2), human review of flagged hosts, monitoring of recall, and re-validation on real assessment data before wider roll-out. Continue the hardening programme, starting with high-port, old-patch Windows hosts.

7. EXPECTED DELIVERABLES AND OUTCOMES SUMMARY

Task	Marks	Expected deliverable (per assignment)	Provided in this report
A1	7	Requirements, plot selection, workflow, model design, R procedure, interpretation and saving	A1.1–A1.6 tables, workflow and steps
A2	5	Diagnosis of weak workflow, redesigned visualizations, model per objective, two improvement actions	A2.1–A2.4
A3	5	Completed visualization table, plot-vs-alternative evaluation, Base R/ggplot2/lattice comparison, 3D/interactive evaluation	A3 table, A3.1–A3.3
A4	4	Model-to-objective table, logistic vs tree criteria, overfitting remedy, accuracy inquiry	A4 table, A4.1–A4.3
A5	4	SLR explanation, structured SLR/MLR/tree selection with R code, performance vs appropriateness	A5.1–A5.3 with R code and results
B1	7	R code, dataset preparation, five labelled graphs, Base R vs ggplot2 comparison, interpretation	Code, 7 figures, screenshots, comparison table, findings
B2	5	Prediction, two strategies, comparison table, parameter changes, 3D/interactive case, recommendation	Prediction, Figures 8–9, tables, recommendation
B3	8	SLR/MLR/logistic/Poisson models, correct metrics, evidence table, reliability analysis	Model screenshots, metric table, ROC/tree, diagnostics
B4	5	End-to-end workflow, misleading-result case with R fix, decision note	Workflow table, threshold analysis, t-test, decision note

Key results at a glance

Analysis	Main result
Hosts / split	400 hosts; 280 training and 120 test
Severity (SLR)	Adjusted R^2 0.153; test RMSE 1.261
Severity (MLR)	Adjusted R^2 0.558; test RMSE 0.971; test R^2 0.510
Compromise (logistic)	Test AUC 0.755; accuracy 0.767; recall 0.367 at 0.5 and 0.800 at 0.15
Compromise (tree)	Test AUC 0.641; recall 0.133
Vulnerability count (Poisson)	AIC 992.5; dispersion 1.13; test RMSE 1.407
Hardening programme	Mean severity 2.75 → 1.99; $p < 0.001$

8. CONCLUSION

The assignment shows that the value of data visualization and predictive modelling in security lies in choosing the right tool for each question. Histograms, box plots, scatter plots, line charts and heatmaps matched to variable types revealed that patch age, open ports and a low hardening score drive severity and vulnerabilities, that Windows hosts carry the highest and most extreme vulnerability counts, and that vulnerabilities fell after the hardening programme began. Multiple linear regression, logistic regression and Poisson regression, each evaluated with its own metric, quantified those effects, while reliability checks exposed overfitting in the decision tree, multicollinearity between Open_Ports and Service_Count, and the danger of accuracy under class imbalance. The recommended decision is to continue the hardening programme and to deploy the logistic model only as a recall-tuned decision-support flag, validated on real assessment data.

APPENDIX A – DATASET DICTIONARY AND COMPLETE R SCRIPT

Column	Description
host_id	Unique host identifier (H001–H400)
Scan_Date	Date of the authorised scan (2025)
OS_Type	Windows, Ubuntu, CentOS or macOS
Open_Ports	Number of open network ports
Patch_Age	Days since last patch (two invalid negative values inserted for cleaning)
Hardening_Score	Benchmark hardening score 0–100 (12 missing values inserted for cleaning)
Failed_Logins	Failed login attempts observed
Service_Count	Number of running services (highly correlated with Open_Ports)
Hardening_Programme	0 = before 1 July 2025, 1 = after
Vulnerability_Count	Vulnerabilities found on the host
Severity_Score	Overall severity 0–10
Compromised	Yes/No

Complete R script (run in order in RStudio; all libraries are on CRAN):

```
install.packages(c('ggplot2','corrplot','car','rpart','rpart.plot','pROC','plotly','lmtest','caret','glmnet'))

# ---- B1.1 Prepare and explore ------------------------------------------
library(ggplot2); library(corrplot); library(car); library(rpart)
library(rpart.plot); library(pROC); library(plotly)
d <- read.csv("vuln_assessment.csv", stringsAsFactors = FALSE)
str(d); summary(d); colSums(is.na(d))          # structure, ranges, missing values
d$OS_Type     <- factor(d$OS_Type)             # categorical -> factor
d$Compromised <- factor(d$Compromised, levels = c("No", "Yes"))
d$Scan_Date   <- as.Date(d$Scan_Date)          # character -> Date
d$Month       <- as.Date(format(d$Scan_Date, "%Y-%m-01"))
sum(d$Patch_Age < 0)                           # inconsistent (impossible) values
d$Patch_Age[d$Patch_Age < 0] <- median(d$Patch_Age[d$Patch_Age >= 0])
d$Hardening_Score[is.na(d$Hardening_Score)] <- median(d$Hardening_Score,
```

```
na.rm = TRUE)
num_vars <- c("Open_Ports","Patch_Age","Hardening_Score","Failed_Logins",
              "Service_Count","Vulnerability_Count","Severity_Score")

# ---- B1.2 Visualization set (Base R + corrplot) ----------------------------
png("f1_hist.png", 1100, 700, res = 160)
hist(d$Severity_Score, breaks = 12, col = "grey", main = "Severity distribution",
     xlab = "Severity score (0-10)", ylab = "Frequency (hosts)")
abline(v = mean(d$Severity_Score), col = "red", lty = 2); dev.off()

png("f2_box_base.png", 1100, 700, res = 160)
boxplot(Vulnerability_Count ~ OS_Type, data = d, main = "Vulnerabilities by OS",
        xlab = "OS type", ylab = "Vulnerability count"); dev.off()

png("f3_scatter.png", 1100, 700, res = 160)
plot(d$Patch_Age, d$Vulnerability_Count, pch = 19, col = "#4C72B0",
     main = "Patch age vs vulnerabilities", xlab = "Patch age (days)", ylab =
"Vulnerability count")
abline(lm(Vulnerability_Count ~ Patch_Age, data = d), col = "red", lwd = 2);
dev.off()

agg <- aggregate(Vulnerability_Count ~ Month, data = d, FUN = sum)
png("f4_line.png", 1100, 700, res = 160)
plot(agg$Month, agg$Vulnerability_Count, type = "b", col = "firebrick",
     main = "Monthly vulnerability trend", xlab = "Month", ylab = "Total
vulnerabilities")
abline(v = as.Date("2025-07-01"), lty = 2, col = "darkgreen"); dev.off()

png("f5_heat.png", 1100, 900, res = 160)
corrplot(cor(d[, num_vars]), method = "color", addCoef.col = "black",
         tl.col = "black", number.cex = 0.7, title = "Correlation heatmap", mar =
c(0,0,2,0)); dev.off()

par(mfrow = c(1, 2))
barplot(table(d$OS_Type), col = 2:5, main = "Hosts per OS", ylab = "Hosts")
pie(table(d$Compromised), main = "Compromised share")   # single share-of-total
only

# ---- B1.3 Base R vs ggplot2 (same question) --------------------------------
boxplot(Vulnerability_Count ~ OS_Type, data = d, main = "Vulns by OS")      # Base
R
```

```
p <- ggplot(d, aes(OS_Type, Vulnerability_Count, fill = OS_Type)) +
    geom_boxplot() + labs(title = "Vulns by OS", x = "OS type", y = "Vulnerability
count") +
    theme_grey()
ggsave("f7_gg_box.png", p, width = 7, height = 4.5, dpi = 300)

# ---- B2 Two strategies for "Do vulnerabilities differ across OS types?" ------
# Strategy A : Base R bar of means
barplot(tapply(d$Vulnerability_Count, d$OS_Type, mean), main = "Mean vulns by
OS",
       ylab = "Mean vulnerability count", col = "grey")
# Strategy B : ggplot2 boxplot + jitter
ggplot(d, aes(OS_Type, Vulnerability_Count, fill = OS_Type)) +
  geom_boxplot(outlier.colour = "red") + geom_jitter(width = 0.1, alpha = 0.3, size =
0.8)
# Parameter changes (B2.2): faceting + log transformation
ggplot(d, aes(OS_Type, Vulnerability_Count + 1, fill = OS_Type)) + geom_boxplot()
+
  facet_wrap(~ Compromised) + scale_y_log10() + labs(y = "Vulnerability count + 1
(log10)")
# Interactive (B2.3)
plotly::ggplotly(p)

# ---- B3.1 Build the models (70/30 split) ----------------------------------
set.seed(5)
i     <- sample(nrow(d), 0.7 * nrow(d)); train <- d[i, ]; test <- d[-i, ]
slr   <- lm(Severity_Score ~ Patch_Age, data = train)
mlr   <- lm(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins, data = train)
logit <- glm(Compromised ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins,
            family = binomial, data = train)
pois  <- glm(Vulnerability_Count ~ Patch_Age + Open_Ports + Hardening_Score,
            family = poisson, data = train)
tree  <- rpart(Compromised ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins,
              data = train, method = "class",
              control = rpart.control(maxdepth = 3, minbucket = 10, cp = 0.01))
summary(slr); summary(mlr); summary(logit); summary(pois)
exp(coef(logit)); exp(coef(pois))                # odds ratios / rate ratios
rpart.plot(tree)
```

```
# ---- B3.2 Performance, a different measure per model type --------------------
rmse <- function(p, a) sqrt(mean((p - a)^2)); mae <- function(p, a) mean(abs(p - a))
summary(slr)$adj.r.squared; summary(mlr)$adj.r.squared
rmse(predict(slr, test), test$Severity_Score); rmse(predict(mlr, test),
test$Severity_Score)
mae(predict(mlr, test), test$Severity_Score)
pr  <- predict(logit, test, type = "response")
cls <- ifelse(pr > 0.5, "Yes", "No")
table(Predicted = cls, Actual = test$Compromised)            # confusion matrix
auc(roc(test$Compromised, pr))                               # AUC
AIC(pois); deviance(pois); sum(residuals(pois, "pearson")^2) / df.residual(pois)  #
dispersion
rmse(predict(pois, test, type = "response"), test$Vulnerability_Count)

# ---- B3.3 Logistic regression vs decision tree on the same test set ----------
pt <- predict(tree, test, type = "prob")[, "Yes"]
auc(roc(test$Compromised, pt))
table(Predicted = ifelse(pt > 0.5, "Yes", "No"), Actual = test$Compromised)

# ---- B3.4 Reliability ---------------------------------------------------------
par(mfrow = c(2, 2)); plot(mlr)                              # residual diagnostics
vif(mlr); vif(update(mlr, . ~ . + Service_Count))            # multicollinearity
library(lmtest); bptest(mlr); shapiro.test(resid(mlr))
which(cooks.distance(mlr) > 4 / nrow(train))                 # influential hosts

# ---- B4 End-to-end decision workflow ------------------------------------------
# (data preparation, graphs and models as above)
t.test(Severity_Score ~ Hardening_Programme, data = d)       # did the programme
reduce severity?
summary(lm(Severity_Score ~ Hardening_Programme + Patch_Age + Open_Ports +
Failed_Logins, data = d))
table(train$Compromised) / nrow(train)                       # class imbalance check
# B4.2 Misleading accuracy and the fix: move the decision threshold
for (th in c(0.5, 0.3, 0.2, 0.15)) {
  cls <- ifelse(pr > th, "Yes", "No")
  cm  <- table(factor(cls, c("No","Yes")), factor(test$Compromised, c("No","Yes")))
  cat(th, "recall =", cm["Yes","Yes"] / sum(cm[, "Yes"]),
          "precision =", cm["Yes","Yes"] / sum(cm["Yes", ]), "\n")
}
# Cross-validated check before deployment
```

```
library(caret)
cv <- train(Compromised ~ Patch_Age + Open_Ports + Hardening_Score +
Failed_Logins, data = d,
          method = "glm", family = binomial, metric = "ROC",
          trControl = trainControl(method = "cv", number = 10, classProbs = TRUE,
                                   summaryFunction = twoClassSummary))
cv

# ---- A5 model-selection procedure ----
set.seed(5); i <- sample(nrow(d), 0.7 * nrow(d)); train <- d[i, ]; test <- d[-i, ]
m1 <- lm(Severity_Score ~ Patch_Age, train)                          # simple
m2 <- lm(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score +
        Failed_Logins + Service_Count, train)                        # multiple
m3 <- rpart(Severity_Score ~ Patch_Age + Open_Ports + Hardening_Score +
          Failed_Logins + Service_Count, train, method = "anova")      # tree
plot(m2); vif(m2); shapiro.test(resid(m2)); library(lmtest); bptest(m2)  # assumptions
AIC(m1, m2); anova(m1, m2)                                           # nested comparison
rmse <- function(p, a) sqrt(mean((p - a)^2))
sapply(list(SLR = m1, MLR = m2, Tree = m3), function(m) rmse(predict(m, test),
test$Severity_Score))
```